7 Vectors

EXERCISES 7.1

Vectors in 2-Space

1. **(a)** $6\mathbf{i}+12\mathbf{j}$ **(b)** $\mathbf{i}+8\mathbf{j}$ **(c)** $3\mathbf{i}$ **(d)** $\sqrt{65}$ **(e)** 3

2. **(a)** $\langle 3,3\rangle$ **(b)** $\langle 3,4\rangle$ **(c)** $\langle -1,-2\rangle$ **(d)** 5 **(e)** $\sqrt{5}$

3. **(a)** $\langle 12,0\rangle$ **(b)** $\langle 4,-5\rangle$ **(c)** $\langle 4,5\rangle$ **(d)** $\sqrt{41}$ **(e)** $\sqrt{41}$

4. **(a)** $\frac{1}{2}\mathbf{i}-\frac{1}{2}\mathbf{j}$ **(b)** $\frac{2}{3}\mathbf{i}+\frac{2}{3}\mathbf{j}$ **(c)** $-\frac{1}{3}\mathbf{i}-\mathbf{j}$ **(d)** $2\sqrt{2}/3$ **(e)** $\sqrt{10}/3$

5. **(a)** $-9\mathbf{i}+6\mathbf{j}$ **(b)** $-3\mathbf{i}+9\mathbf{j}$ **(c)** $-3\mathbf{i}-5\mathbf{j}$ **(d)** $3\sqrt{10}$ **(e)** $\sqrt{34}$

6. **(a)** $\langle 3,9\rangle$ **(b)** $\langle -4,-12\rangle$ **(c)** $\langle 6,18\rangle$ **(d)** $4\sqrt{10}$ **(e)** $6\sqrt{10}$

7. **(a)** $-6\mathbf{i}+27\mathbf{j}$ **(b)** $\mathbf{0}$ **(c)** $-4\mathbf{i}+18\mathbf{j}$ **(d)** 0 **(e)** $2\sqrt{85}$

8. **(a)** $\langle 21,30\rangle$ **(b)** $\langle 8,12\rangle$ **(c)** $\langle 6,8\rangle$ **(d)** $4\sqrt{13}$ **(e)** 10

9. **(a)** $\langle 4,-12\rangle-\langle -2,2\rangle=\langle 6,-14\rangle$ **(b)** $\langle -3,9\rangle-\langle -5,5\rangle=\langle 2,4\rangle$

10. **(a)** $(4\mathbf{i}+4\mathbf{j})-(6\mathbf{i}-4\mathbf{j})=-2\mathbf{i}+8\mathbf{j}$ **(b)** $(-3\mathbf{i}-3\mathbf{j})-(15\mathbf{i}-10\mathbf{j})=-18\mathbf{i}+7\mathbf{j}$

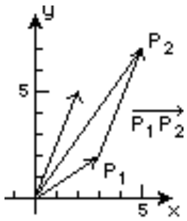
11. **(a)** $(4\mathbf{i}-4\mathbf{j})-(-6\mathbf{i}+8\mathbf{j})=10\mathbf{i}-12\mathbf{j}$ **(b)** $(-3\mathbf{i}+3\mathbf{j})-(-15\mathbf{i}+20\mathbf{j})=12\mathbf{i}-17\mathbf{j}$

12. **(a)** $\langle 8,0\rangle-\langle 0,-6\rangle=\langle 8,6\rangle$ **(b)** $\langle -6,0\rangle-\langle 0,-15\rangle=\langle -6,15\rangle$

13. **(a)** $\langle 16,40\rangle-\langle -4,-12\rangle=\langle 20,52\rangle$ **(b)** $\langle -12,-30\rangle-\langle -10,-30\rangle=\langle -2,0\rangle$

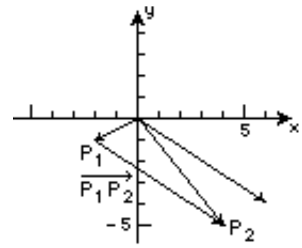
14. **(a)** $\langle 8,12\rangle-\langle 10,6\rangle=\langle -2,6\rangle$ **(b)** $\langle -6,-9\rangle-\langle 25,15\rangle=\langle -31,-24\rangle$

15.



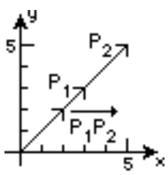
$\overrightarrow{P_1P_2} = \langle 2, 5 \rangle$

16.



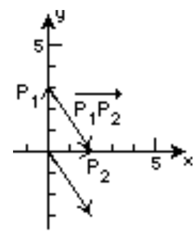
$\overrightarrow{P_1P_2} = \langle 6, -4 \rangle$

17.



$\overrightarrow{P_1P_2} = \langle 2, 2 \rangle$

18.



$\overrightarrow{P_1P_2} = \langle 2, -3 \rangle$

19. Since $\overrightarrow{P_1P_2} = \overrightarrow{OP_2} - \overrightarrow{OP_1}$, $\overrightarrow{OP_2} = \overrightarrow{P_1P_2} + \overrightarrow{OP_1} = (4\mathbf{i} + 8\mathbf{j}) + (-3\mathbf{i} + 10\mathbf{j}) = \mathbf{i} + 18\mathbf{j}$, and the terminal point is $(1, 18)$.

20. Since $\overrightarrow{P_1P_2} = \overrightarrow{OP_2} - \overrightarrow{OP_1}$, $\overrightarrow{OP_1} = \overrightarrow{OP_2} - \overrightarrow{P_1P_2} = \langle 4, 7 \rangle - \langle -5, -1 \rangle = \langle 9, 8 \rangle$, and the initial point is $(9, 8)$.

21. $a(= -\mathbf{a})$, $b(= -\frac{1}{4}\mathbf{a})$, $c(= \frac{5}{2}\mathbf{a})$, $e(= 2\mathbf{a})$, and $f(= -\frac{1}{2}\mathbf{a})$ are parallel to $\mathbf{a}$.

22. We want $-3\mathbf{b} = \mathbf{a}$, so $c = -3(9) = -27$.

23. $\langle 6, 15 \rangle$

24. $\langle 5, 2 \rangle$

25. $\|\mathbf{a}\| = \sqrt{4+4} = 2\sqrt{2}$; **(a)** $\mathbf{u} = \frac{1}{2\sqrt{2}}\langle 2, 2 \rangle = \langle \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \rangle$; **(b)** $-\mathbf{u} = \langle -\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}} \rangle$

26. $\|\mathbf{a}\| = \sqrt{9+16} = 5$; **(a)** $\mathbf{u} = \frac{1}{5}\langle -3, 4 \rangle = \langle -\frac{3}{5}, \frac{4}{5} \rangle$; **(b)** $-\mathbf{u} = \langle \frac{3}{5}, -\frac{4}{5} \rangle$

27. $\|\mathbf{a}\| = 5$; **(a)** $\mathbf{u} = \frac{1}{5}\langle 0, -5 \rangle = \langle 0, -1 \rangle$; **(b)** $-\mathbf{u} = \langle 0, 1 \rangle$

28. $\|\mathbf{a}\| = \sqrt{1+3} = 2$; **(a)** $\mathbf{u} = \frac{1}{2}\langle 1, -\sqrt{3} \rangle = \langle \frac{1}{2}, -\frac{\sqrt{3}}{2} \rangle$; **(b)** $-\mathbf{u} = \langle -\frac{1}{2}, \frac{\sqrt{3}}{2} \rangle$

29. $\|\mathbf{a} + \mathbf{b}\| = \|\langle 5, 12 \rangle\| = \sqrt{25+144} = 13$; $\mathbf{u} = \frac{1}{13}\langle 5, 12 \rangle = \langle \frac{5}{13}, \frac{12}{13} \rangle$

30. $\|2\mathbf{a} - 3\mathbf{b}\| = \|\langle -5, 4 \rangle\| = \sqrt{25+16} = \sqrt{41}$; $\mathbf{u} = \frac{1}{\sqrt{41}}\langle -5, 4 \rangle = \langle -\frac{5}{\sqrt{41}}, \frac{4}{\sqrt{41}} \rangle$

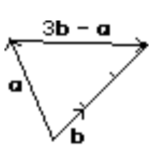
31. $\|\mathbf{a}\| = \sqrt{9+49} = \sqrt{58}$; $\mathbf{b} = 2(\frac{1}{\sqrt{58}})(3\mathbf{i} + 7\mathbf{j}) = \frac{6}{\sqrt{58}}\mathbf{i} + \frac{14}{\sqrt{58}}\mathbf{j}$

32. $\|\mathbf{a}\| = \sqrt{\frac{1}{4} + \frac{1}{4}} = \frac{1}{\sqrt{2}}$; $\mathbf{b} = 3(\frac{1}{1/\sqrt{2}})(\frac{1}{2}\mathbf{i} - \frac{1}{2}\mathbf{j}) = \frac{3\sqrt{2}}{2}\mathbf{i} - \frac{3\sqrt{2}}{2}\mathbf{j}$

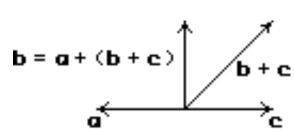
33. $-\frac{3}{4}\mathbf{a} = \langle -3, -15/2 \rangle$

34. $5(\mathbf{a} + \mathbf{b}) = 5\langle 0, 1 \rangle = \langle 0, 5 \rangle$

35.

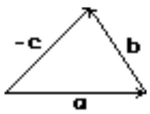


36.

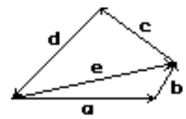


37. $\mathbf{x} = -(\mathbf{a} + \mathbf{b}) = -\mathbf{a} - \mathbf{b}$

38. $\mathbf{x} = 2(\mathbf{a} - \mathbf{b}) = 2\mathbf{a} - 2\mathbf{b}$

39. 

$\mathbf{b} = (-\mathbf{c}) - \mathbf{a}$; $(\mathbf{b}+\mathbf{c})+\mathbf{a} = \mathbf{0}$; $\mathbf{a}+\mathbf{b}+\mathbf{c} = \mathbf{0}$

40. 

From Problem 39, $\mathbf{e}+\mathbf{c}+\mathbf{d} = \mathbf{0}$. But $\mathbf{b} = \mathbf{e}-\mathbf{a}$ and $\mathbf{e} = \mathbf{a}+\mathbf{b}$, so $(\mathbf{a}+\mathbf{b})+\mathbf{c}+\mathbf{d} = \mathbf{0}$.

41. From $2\mathbf{i}+3\mathbf{j} = k_1\mathbf{b}+k_2\mathbf{c} = k_1(\mathbf{i}+\mathbf{j})+k_2(\mathbf{i}-\mathbf{j}) = (k_1+k_2)\mathbf{i}+(k_1-k_2)\mathbf{j}$ we obtain the system of equations $k_1+k_2 = 2$, $k_1-k_2 = 3$. Solving, we find $k_1 = \frac{5}{2}$ and $k_2 = -\frac{1}{2}$. Then $\mathbf{a} = \frac{5}{2}\mathbf{b} - \frac{1}{2}\mathbf{c}$.

42. From $2\mathbf{i}+3\mathbf{j} = k_1\mathbf{b}+k_2\mathbf{c} = k_1(-2\mathbf{i}+4\mathbf{j})+k_2(5\mathbf{i}+7\mathbf{j}) = (-2k_1+5k_2)\mathbf{i}+(4k_1+7k_2)\mathbf{j}$ we obtain the system of equations $-2k_1+5k_2 = 2$, $4k_1+7k_2 = 3$. Solving, we find $k_1 = \frac{1}{34}$ and $k_2 = \frac{7}{17}$.

43. From $y' = \frac{1}{2}x$ we see that the slope of the tangent line at $(2,2)$ is 1. A vector with slope 1 is $\mathbf{i}+\mathbf{j}$. A unit vector is $(\mathbf{i}+\mathbf{j})/\|\mathbf{i}+\mathbf{j}\| = (\mathbf{i}+\mathbf{j})/\sqrt{2} = \frac{1}{\sqrt{2}}\mathbf{i}+\frac{1}{\sqrt{2}}\mathbf{j}$. Another unit vector tangent to the curve is $-\frac{1}{\sqrt{2}}\mathbf{i}-\frac{1}{\sqrt{2}}\mathbf{j}$.

44. From $y' = -2x+3$ we see that the slope of the tangent line at $(0,0)$ is 3. A vector with slope 3 is $\mathbf{i}+3\mathbf{j}$. A unit vector is $(\mathbf{i}+3\mathbf{j})/\|\mathbf{i}+3\mathbf{j}\| = (\mathbf{i}+3\mathbf{j})/\sqrt{10} = \frac{1}{\sqrt{10}}\mathbf{i}+\frac{3}{\sqrt{10}}\mathbf{j}$. Another unit vector is $-\frac{1}{\sqrt{10}}\mathbf{i}-\frac{3}{\sqrt{10}}\mathbf{j}$.

45. **(a)** Since $\mathbf{F}_f = -\mathbf{F}_g$, $\|\mathbf{F}_g\| = \|\mathbf{F}_f\| = \mu\|\mathbf{F}_n\|$ and $\tan\theta = \|\mathbf{F}_g\|/\|\mathbf{F}_n\| = \mu\|\mathbf{F}_n\|/\|\mathbf{F}_n\| = \mu$.

(b) $\theta = \tan^{-1} 0.6 \approx 31°$

46. Since $\mathbf{w}+\mathbf{F}_1+\mathbf{F}_2 = \mathbf{0}$,

$$-200\mathbf{j}+\|\mathbf{F}_1\|\cos 20°\mathbf{i}+\|\mathbf{F}_1\|\sin 20°\mathbf{j}-\|\mathbf{F}_2\|\cos 15°\mathbf{i}+\|\mathbf{F}_2\|\sin 15°\mathbf{j} = \mathbf{0}$$

or

$$(\|\mathbf{F}_1\|\cos 20° - \|\mathbf{F}_2\|\cos 15°)\mathbf{i}+(\|\mathbf{F}_1\|\sin 20°+\|\mathbf{F}_2\|\sin 15° - 200)\mathbf{j} = \mathbf{0}.$$

Thus, $\|\mathbf{F}_1\|\cos 20° - \|\mathbf{F}_2\|\cos 15° = 0$; $\|\mathbf{F}_1\|\sin 20°+\|\mathbf{F}_2\|\sin 15° - 200 = 0$. Solving this system for $\|\mathbf{F}_1\|$ and $\|\mathbf{F}_2\|$, we obtain

$$\|\mathbf{F}_1\| = \frac{200\cos 15°}{\sin 15°\cos 20°+\cos 15°\sin 20°} = \frac{200\cos 15°}{\sin(15°+20°)} = \frac{200\cos 15°}{\sin 35°} \approx 336.8 \text{ lb}$$

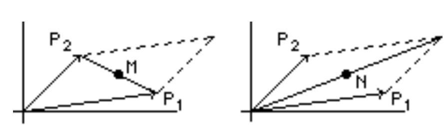
and

$$\|\mathbf{F}_2\| = \frac{200\cos 20°}{\sin 15°\cos 20°+\cos 15°\sin 20°} = \frac{200\cos 20°}{\sin 35°} \approx 327.7 \text{ lb}.$$

47. Since $y/2a(L^2+y^2)^{3/2}$ is an odd function on $[-a,a]$, $F_y = 0$. Now, using the fact that $L/(L^2+y^2)^{3/2}$ is an even function, we have

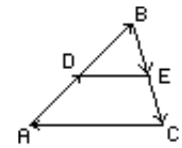
$$\begin{aligned}
\int_{-a}^{a}\frac{L\,dy}{2a(L^2+y^2)^{3/2}} &= \frac{L}{a}\int_0^a \frac{dy}{(L^2+y^2)^{3/2}} \qquad \boxed{y = L\tan\theta,\ dy = L\sec^2\theta\,d\theta} \\
&= \frac{L}{a}\int_0^{\tan^{-1}a/L}\frac{L\sec^2\theta\,d\theta}{L^3(1+\tan^2\theta)^{3/2}} = \frac{1}{La}\int_0^{\tan^{-1}a/L}\frac{\sec^2\theta\,d\theta}{\sec^3\theta} \\
&= \frac{1}{La}\int_0^{\tan^{-1}a/L}\cos\theta\,d\theta = \frac{1}{La}\sin\theta\Big|_0^{\tan^{-1}a/L} \\
&= \frac{1}{La}\frac{a}{\sqrt{L^2+a^2}} = \frac{1}{L\sqrt{L^2+a^2}}.
\end{aligned}$$

Then $F_x = qQ/4\pi\epsilon_0 L\sqrt{L^2+a^2}$ and $\mathbf{F} = (qQ/4\pi\epsilon_0 L\sqrt{L^2+a^2}\,)\mathbf{i}$.

48. 

Place one corner of the parallelogram at the origin and let two adjacent sides be $\overrightarrow{OP_1}$ and $\overrightarrow{OP_2}$. Let M be the midpoint of the diagonal connecting P_1 and P_2 and N be the midpoint of the other diagonal. Then $\overrightarrow{OM} = \frac{1}{2}(\overrightarrow{OP_1}+\overrightarrow{OP_2})$. Since $\overrightarrow{OP_1}+\overrightarrow{OP_2}$ is the main diagonal of the parallelogram and N is its midpoint, $\overrightarrow{ON} = \frac{1}{2}(\overrightarrow{OP_1}+\overrightarrow{OP_2})$. Thus, $\overrightarrow{OM} = \overrightarrow{ON}$ and the diagonals bisect each other.

49. By Problem 39, $\overrightarrow{AB}+\overrightarrow{BC}+\overrightarrow{CA}=\mathbf{0}$ and $\overrightarrow{AD}+\overrightarrow{DE}+\overrightarrow{EC}+\overrightarrow{CA}=\mathbf{0}$. From the first equation, $\overrightarrow{AB}+\overrightarrow{BC}=-\overrightarrow{CA}$. Since D and E are midpoints, $\overrightarrow{AD}=\frac{1}{2}\overrightarrow{AB}$ and $\overrightarrow{EC}=\frac{1}{2}\overrightarrow{BC}$. Then, $\frac{1}{2}\overrightarrow{AB}+\overrightarrow{DE}+\frac{1}{2}\overrightarrow{BC}+\overrightarrow{CA}=\mathbf{0}$ and



$$\overrightarrow{DE}=-\overrightarrow{CA}-\frac{1}{2}(\overrightarrow{AB}+\overrightarrow{BC})=-\overrightarrow{CA}-\frac{1}{2}(-\overrightarrow{CA})=-\frac{1}{2}\overrightarrow{CA}.$$

Thus, the line segment joining the midpoints D and E is parallel to the side AC and half its length.

50. We have $\overrightarrow{OA}=150\cos 20°\mathbf{i}+150\sin 20°\mathbf{j}$, $\overrightarrow{AB}=200\cos 113°\mathbf{i}+200\sin 113°\mathbf{j}$, $\overrightarrow{BC}=240\cos 190°\mathbf{i}+240\sin 190°\mathbf{j}$. Then

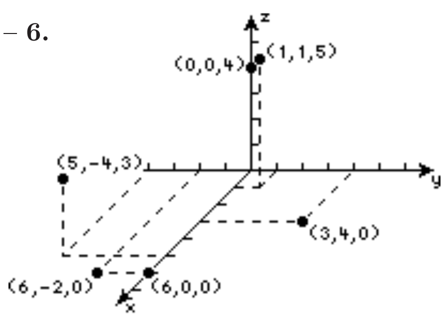
$$\begin{aligned}\mathbf{r}&=(150\cos 20°+200\cos 113°+240\cos 190°)\mathbf{i}+(150\sin 20°+200\sin 113°+240\sin 190°)\mathbf{j}\\&\approx -173.55\mathbf{i}+193.73\mathbf{j}\end{aligned}$$

and $\|\mathbf{r}\|\approx 260.09$ miles.

EXERCISES 7.2

Vectors in 3-Space

1. – 6.



7. A plane perpendicular to the z-axis, 5 units above the xy-plane

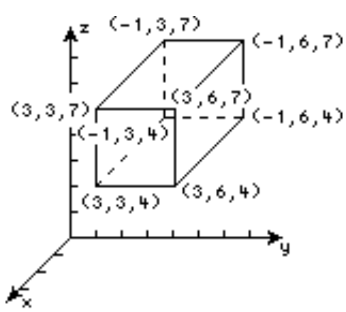
8. A plane perpendicular to the x-axis, 1 unit in front of the yz-plane

9. A line perpendicular to the xy-plane at $(2,3,0)$

10. A single point located at $(4,-1,7)$

11. $(2,0,0)$, $(2,5,0)$, $(2,0,8)$, $(2,5,8)$, $(0,5,0)$, $(0,5,8)$, $(0,0,8)$, $(0,0,0)$

12.



13. **(a)** xy-plane: $(-2,5,0)$, xz-plane: $(-2,0,4)$, yz-plane: $(0,5,4)$; **(b)** $(-2,5,-2)$

(c) Since the shortest distance between a point and a plane is a perpendicular line, the point in the plane $x = 3$ is $(3, 5, 4)$.

14. We find planes that are parallel to coordinate planes: **(a)** $z = -5$; **(b)** $x = 1$ and $y = -1$; **(c)** $z = 2$

15. The union of the planes $x = 0$, $y = 0$, and $z = 0$

16. The origin $(0, 0, 0)$

17. The point $(-1, 2, -3)$

18. The union of the planes $x = 2$ and $z = 8$

19. The union of the planes $z = 5$ and $z = -5$

20. The line through the points $(1, 1, 1)$, $(-1, -1, -1)$, and the origin

21. $d = \sqrt{(3-6)^2 + (-1-4)^2 + (2-8)^2} = \sqrt{70}$

22. $d = \sqrt{(-1-0)^2 + (-3-4)^2 + (5-3)^2} = 3\sqrt{6}$

23. **(a)** 7; **(b)** $d = \sqrt{(-3)^2 + (-4)^2} = 5$

24. **(a)** 2; **(b)** $d = \sqrt{(-6)^2 + 2^2 + (-3)^2} = 7$

25. $d(P_1, P_2) = \sqrt{3^2 + 6^2 + (-6)^2} = 9$; $d(P_1, P_3) = \sqrt{2^2 + 1^2 + 2^2} = 3$
$d(P_2, P_3) = \sqrt{(2-3)^2 + (1-6)^2 + (2-(-6))^2} = \sqrt{90}$; The triangle is a right triangle.

26. $d(P_1, P_2) = \sqrt{1^2 + 2^2 + 4^2} = \sqrt{21}$; $d(P_1, P_3) = \sqrt{3^2 + 2^2 + (2\sqrt{2})^2} = \sqrt{21}$
$d(P_2, P_3) = \sqrt{(3-1)^2 + (2-2)^2 + (2\sqrt{2}-4)^2} = \sqrt{28 - 16\sqrt{2}}$
The triangle is an isosceles triangle.

27. $d(P_1, P_2) = \sqrt{(4-1)^2 + (1-2)^2 + (3-3)^2} = \sqrt{10}$
$d(P_1, P_3) = \sqrt{(4-1)^2 + (6-2)^2 + (4-3)^2} = \sqrt{26}$
$d(P_2, P_3) = \sqrt{(4-4)^2 + (6-1)^2 + (4-3)^2} = \sqrt{26}$; The triangle is an isosceles triangle.

28. $d(P_1, P_2) = \sqrt{(1-1)^2 + (1-1)^2 + (1-(-1))^2} = 2$
$d(P_1, P_3) = \sqrt{(0-1)^2 + (-1-1)^2 + (1-(-1))^2} = 3$
$d(P_2, P_3) = \sqrt{(0-1)^2 + (-1-1)^2 + (1-1)^2} = \sqrt{5}$; The triangle is a right triangle.

29. $d(P_1, P_2) = \sqrt{(-2-1)^2 + (-2-2)^2 + (-3-0)^2} = \sqrt{34}$
$d(P_1, P_3) = \sqrt{(7-1)^2 + (10-2)^2 + (6-0)^2} = 2\sqrt{34}$
$d(P_2, P_3) = \sqrt{(7-(-2))^2 + (10-(-2))^2 + (6-(-3))^2} = 3\sqrt{34}$
Since $d(P_1, P_2) + d(P_1, P_3) = d(P_2, P_3)$, the points P_1, P_2, and P_3 are collinear.

30. $d(P_1, P_2) = \sqrt{(1-2)^2 + (4-3)^2 + (4-2)^2} = \sqrt{6}$
$d(P_1, P_3) = \sqrt{(5-2)^2 + (0-3)^2 + (-4-2)^2} = 3\sqrt{6}$
$d(P_2, P_3) = \sqrt{(5-1)^2 + (0-4)^2 + (-4-4)^2} = 4\sqrt{6}$
Since $d(P_1, P_2) + d(P_1, P_3) = d(P_2, P_3)$, the points P_1, P_2, and P_3 are collinear.

31. $\sqrt{(2-x)^2 + (1-2)^2 + (1-3)^2} = \sqrt{21} \implies x^2 - 4x + 9 = 21 \implies x^2 - 4x + 4 = 16$
$\implies (x-2)^2 = 16 \implies x = 2 \pm 4$ or $x = 6, -2$

32. $\sqrt{(0-x)^2+(3-x)^2+(5-1)^2}=5 \implies 2x^2-6x+25=25 \implies x^2-3x=0 \implies x=0,3$

33. $\left(\dfrac{1+7}{2},\dfrac{3+(-2)}{2},\dfrac{1/2+5/2}{2}\right)=(4,1/2,3/2)$

34. $\left(\dfrac{0+4}{2},\dfrac{5+1}{2},\dfrac{-8+(-6)}{2}\right)=(2,3,-7)$

35. $(x_1+2)/2=-1,\ x_1=-4;\quad (y_1+3)/2=-4,\ y_1=-11;\quad (z_1+6)/2=8,\quad z_1=10$
The coordinates of P_1 are $(-4,-11,10)$.

36. $(-3+(-5))/2=x_3=-4;\quad (4+8)/2=y_3=6;\quad (1+3)/2=z_3=2.$
The coordinates of P_3 are $(-4,6,2)$.

(a) $\left(\dfrac{-3+(-4)}{2},\dfrac{4+6}{2},\dfrac{1+2}{2}\right)=(-7/2,5,3/2)$

(b) $\left(\dfrac{-4+(-5)}{2},\dfrac{6+8}{2},\dfrac{2+3}{2}\right)=(-9/2,7,5/2)$

37. $\overrightarrow{P_1P_2}=\langle -3,-6,1\rangle$

38. $\overrightarrow{P_1P_2}=\langle 8,-5/2,8\rangle$

39. $\overrightarrow{P_1P_2}=\langle 2,1,1\rangle$

40. $\overrightarrow{P_1P_2}\langle -3,-3,7\rangle$

41. $\mathbf{a}+(\mathbf{b}+\mathbf{c})=\langle 2,4,12\rangle$

42. $2\mathbf{a}-(\mathbf{b}-\mathbf{c})=\langle 2,-6,4\rangle-\langle -3,-5,-8\rangle=\langle 5,-1,12\rangle$

43. $\mathbf{b}+2(\mathbf{a}-3\mathbf{c})=\langle -1,1,1\rangle+2\langle -5,-21,-25\rangle=\langle -11,-41,-49\rangle$

44. $4(\mathbf{a}+2\mathbf{c})-6\mathbf{b}=4\langle 5,9,20\rangle-\langle -6,6,6\rangle=\langle 26,30,74\rangle$

45. $\|\mathbf{a}+\mathbf{c}\|=\|\langle 3,3,11\rangle\|=\sqrt{9+9+121}=\sqrt{139}$

46. $\|\mathbf{c}\|\|2\mathbf{b}\|=(\sqrt{4+36+81})(2)(\sqrt{1+1+1})=22\sqrt{3}$

47. $\left\|\dfrac{\mathbf{a}}{\|\mathbf{a}\|}\right\|+5\left\|\dfrac{\mathbf{b}}{|\mathbf{b}\|}\right\|=\dfrac{1}{\|\mathbf{a}\|}\|\mathbf{a}\|+5\dfrac{1}{\|\mathbf{b}\|}\|\mathbf{b}\|=1+5=6$

48. $\|\mathbf{b}\|\mathbf{a}+\|\mathbf{a}\|\mathbf{b}=\sqrt{1+1+1}\,\langle 1,-3,2\rangle+\sqrt{1+9+4}\,\langle -1,1,1\rangle=\langle\sqrt{3},-3\sqrt{3},2\sqrt{3}\rangle+\langle -\sqrt{14},\sqrt{14},\sqrt{14}\rangle$
$=\langle\sqrt{3}-\sqrt{14},-3\sqrt{3}+\sqrt{14},2\sqrt{3}+\sqrt{14}\rangle$

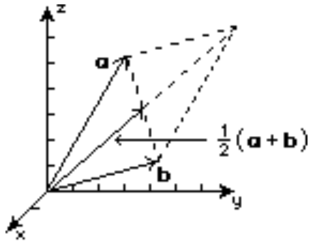
49. $\|\mathbf{a}\|=\sqrt{100+25+100}=15;\quad \mathbf{u}=-\dfrac{1}{15}\langle 10,-5,10\rangle=\langle -2/3,1/3,-2/3\rangle$

50. $\|\mathbf{a}\|=\sqrt{1+9+4}=\sqrt{14};\quad \mathbf{u}=\dfrac{1}{\sqrt{14}}(\mathbf{i}-3\mathbf{j}+2\mathbf{k})=\dfrac{1}{\sqrt{14}}\mathbf{i}-\dfrac{3}{\sqrt{14}}\mathbf{j}+\dfrac{2}{\sqrt{14}}\mathbf{k}$

51. $\mathbf{b}=4\mathbf{a}=4\mathbf{i}-4\mathbf{j}+4\mathbf{k}$

52. $\|\mathbf{a}\|=\sqrt{36+9+4}=7;\quad \mathbf{b}=-\dfrac{1}{2}\left(\dfrac{1}{7}\right)\langle -6,3,-2\rangle=\langle\dfrac{3}{7},-\dfrac{3}{14},\dfrac{1}{7}\rangle$

53.



EXERCISES 7.3

Dot Product

1. $\mathbf{a}\cdot\mathbf{b} = 10(5)\cos(\pi/4) = 25\sqrt{2}$

2. $\mathbf{a}\cdot\mathbf{b} = 6(12)\cos(\pi/6) = 36\sqrt{3}$

3. $\mathbf{a}\cdot\mathbf{b} = 2(-1) + (-3)2 + 4(5) = 12$

4. $\mathbf{b}\cdot\mathbf{c} = (-1)3 + 2(6) + 5(-1) = 4$

5. $\mathbf{a}\cdot\mathbf{c} = 2(3) + (-3)6 + 4(-1) = -16$

6. $\mathbf{a}\cdot(\mathbf{b}+\mathbf{c}) = 2(2) + (-3)8 + 4(4) = -4$

7. $\mathbf{a}\cdot(4\mathbf{b}) = 2(-4) + (-3)8 + 4(20) = 48$

8. $\mathbf{b}\cdot(\mathbf{a}-\mathbf{c}) = (-1)(-1) + 2(-9) + 5(5) = 8$

9. $\mathbf{a}\cdot\mathbf{a} = 2^2 + (-3)^2 + 4^2 = 29$

10. $(2\mathbf{b})\cdot(3\mathbf{c}) = (-2)9 + 4(18) + 10(-3) = 24$

11. $\mathbf{a}\cdot(\mathbf{a}+\mathbf{b}+\mathbf{c}) = 2(4) + (-3)5 + 4(8) = 25$

12. $(2\mathbf{a})\cdot(\mathbf{a}-2\mathbf{b}) = 4(4) + (-6)(-7) + 8(-6) = 10$

13. $\left(\dfrac{\mathbf{a}\cdot\mathbf{b}}{\mathbf{b}\cdot\mathbf{b}}\right)\mathbf{b} = \left[\dfrac{2(-1)+(-3)2+4(5)}{(-1)^2+2^2+5^2}\right]\langle -1,2,5\rangle = \dfrac{12}{30}\langle -1,2,5\rangle = \langle -2/5, 4/5, 2\rangle$

14. $(\mathbf{c}\cdot\mathbf{b})\mathbf{a} = [3(-1) + 6(2) + (-1)5]\langle 2,-3,4\rangle = 4\langle 2,-3,4\rangle = \langle 8,-12,16\rangle$

15. a and f, b and e, c and d

16. **(a)** $\mathbf{a}\cdot\mathbf{b} = 2\cdot 3 + (-c)2 + 3(4) = 0 \implies c = 9$

(b) $\mathbf{a}\cdot\mathbf{b} = c(-3) + \frac{1}{2}(4) + c^2 = c^2 - 3c + 2 = (c-2)(c-1) = 0 \implies c = 1, 2$

17. Solving the system of equations $3x_1 + y_1 - 1 = 0$, $-3x_1 + 2y_1 + 2 = 0$ gives $x_1 = 4/9$ and $y_1 = -1/3$. Thus, $\mathbf{v} = \langle 4/9, -1/3, 1\rangle$.

18. If $\mathbf{a}$ and $\mathbf{b}$ represent adjacent sides of the rhombus, then $\|\mathbf{a}\| = \|\mathbf{b}\|$, the diagonals of the rhombus are $\mathbf{a}+\mathbf{b}$ and $\mathbf{a}-\mathbf{b}$, and

$$(\mathbf{a}+\mathbf{b})\cdot(\mathbf{a}-\mathbf{b}) = \mathbf{a}\cdot\mathbf{a} - \mathbf{a}\cdot\mathbf{b} + \mathbf{b}\cdot\mathbf{a} - \mathbf{b}\cdot\mathbf{b} = \mathbf{a}\cdot\mathbf{a} - \mathbf{b}\cdot\mathbf{b} = \|\mathbf{a}\|^2 - \|\mathbf{b}\|^2 = 0.$$

Thus, the diagonals are perpendicular.

19. Since

$$\mathbf{c}\cdot\mathbf{a} = \left(\mathbf{b} - \frac{\mathbf{a}\cdot\mathbf{b}}{\|\mathbf{a}\|^2}\mathbf{a}\right)\cdot\mathbf{a} = \mathbf{b}\cdot\mathbf{a} - \frac{\mathbf{a}\cdot\mathbf{b}}{\|\mathbf{a}\|^2}(\mathbf{a}\cdot\mathbf{a}) = \mathbf{b}\cdot\mathbf{a} - \frac{\mathbf{a}\cdot\mathbf{b}}{\|\mathbf{a}\|^2}\|\mathbf{a}\|^2 = \mathbf{b}\cdot\mathbf{a} - \mathbf{a}\cdot\mathbf{b} = 0,$$

the vectors $\mathbf{c}$ and $\mathbf{a}$ are orthogonal.

20. $\mathbf{a}\cdot\mathbf{b} = 1(1) + c(1) = c + 1$; $\|\mathbf{a}\| = \sqrt{1+c^2}$, $\|\mathbf{b}\| = \sqrt{2}$

$$\cos 45^\circ = \frac{1}{\sqrt{2}} = \frac{c+1}{\sqrt{1+c^2}\sqrt{2}} \implies \sqrt{1+c^2} = c+1 \implies 1 + c^2 = c^2 + 2c + 1 \implies c = 0$$

21. $\mathbf{a}\cdot\mathbf{b} = 3(2) + (-1)2 = 4$; $\|\mathbf{a}\| = \sqrt{10}$, $\|\mathbf{b}\| = 2\sqrt{2}$

$$\cos\theta = \frac{4}{(\sqrt{10})(2\sqrt{2})} = \frac{1}{\sqrt{5}} \implies \theta = \cos^{-1}\frac{1}{\sqrt{5}} \approx 1.11 \text{ rad} \approx 63.43^\circ$$

22. $\mathbf{a}\cdot\mathbf{b} = 2(-3) + 1(-4) = -10$; $\|\mathbf{a}\| = \sqrt{5}$, $\|\mathbf{b}\| = 5$

$$\cos\theta = \frac{-10}{(\sqrt{5}\,)5} = -\frac{2}{\sqrt{5}} \implies \theta = \cos^{-1}(-2/\sqrt{5}\,) \approx 2.68 \text{ rad} \approx 153.43°$$

23. $\mathbf{a}\cdot\mathbf{b} = 2(-1)+4(-1)+0(4) = -6;\ \ \|\mathbf{a}\| = 2\sqrt{5}\,,\ \ \|\mathbf{b}\| = 3\sqrt{2}$

$$\cos\theta = \frac{-6}{(2\sqrt{5})(3\sqrt{2})} = -\frac{1}{\sqrt{10}} \implies \theta = \cos^{-1}(-1/\sqrt{10}\,) \approx 1.89 \text{ rad} \approx 108.43°$$

24. $\mathbf{a}\cdot\mathbf{b} = \frac{1}{2}(2)+\frac{1}{2}(-4)+\frac{3}{2}(6) = 8;\ \ \|\mathbf{a}\| = \sqrt{11}/2,\ \ \|\mathbf{b}\| = 2\sqrt{14}$

$$\cos\theta = \frac{8}{(\sqrt{11}/2)(2\sqrt{14}\,)} = \frac{8}{\sqrt{154}} \implies \theta = \cos^{-1}(8/\sqrt{154}\,) \approx 0.87 \text{ rad} \approx 49.86°$$

25. $\|\mathbf{a}\| = \sqrt{14}\,;\ \ \cos\alpha = 1/\sqrt{14}\,,\ \ \alpha \approx 74.50°;\ \ \cos\beta = 2/\sqrt{14}\,,\ \ \beta \approx 57.69°;\ \ \cos\gamma = 3/\sqrt{14}\,,\ \ \gamma \approx 36.70°$

26. $\|\mathbf{a}\| = 9;\ \ \cos\alpha = 2/3,\ \ \alpha \approx 48.19°;\ \ \cos\beta = 2/3,\ \ \beta \approx 48.19°;\ \ \cos\gamma = -1/3,\ \ \gamma \approx 109.47°$

27. $\|\mathbf{a}\| = 2;\ \ \cos\alpha = 1/2,\ \ \alpha = 60°;\ \ \cos\beta = 0,\ \ \beta = 90°;\ \ \cos\gamma = -\sqrt{3}/2,\ \ \gamma = 150°$

28. $\|\mathbf{a}\| = \sqrt{78}\,;\ \ \cos\alpha = 5/\sqrt{78}\,,\ \ \alpha \approx 55.52°;\ \ \cos\beta = 7/\sqrt{78}\,,\ \ \beta \approx 37.57°;\ \ \cos\gamma = 2/\sqrt{78}\,,\ \ \gamma \approx 76.91°$

29. Let θ be the angle between $\overrightarrow{AD}$ and $\overrightarrow{AB}$ and a be the length of an edge of the cube. Then $\overrightarrow{AD} = a\mathbf{i}+a\mathbf{j}+a\mathbf{k}$, $\overrightarrow{AB} = a\mathbf{i}$ and

$$\cos\theta = \frac{\overrightarrow{AD}\cdot\overrightarrow{AB}}{\|\overrightarrow{AD}\|\,\|\overrightarrow{AB}\|} = \frac{a^2}{\sqrt{3a^2}\,\sqrt{a^2}} = \frac{1}{\sqrt{3}}$$

so $\theta \approx 0.955317$ radian or $54.7356°$. Letting ϕ be the angle between $\overrightarrow{AD}$ and $\overrightarrow{AC}$ and noting that $\overrightarrow{AC} = a\mathbf{i}+a\mathbf{j}$ we have

$$\cos\phi = \frac{\overrightarrow{AD}\cdot\overrightarrow{AC}}{\|\overrightarrow{AD}\|\,\|\overrightarrow{AC}\|} = \frac{a^2+a^2}{\sqrt{3a^2}\,\sqrt{2a^2}} = \sqrt{\frac{2}{3}}$$

so $\phi \approx 0.61548$ radian or $35.2644°$.

30. If $\mathbf{a}$ and $\mathbf{b}$ are orthogonal, then $\mathbf{a}\cdot\mathbf{b} = 0$ and

$$\begin{aligned}\cos\alpha_1\cos\alpha_2+\cos\beta_1\cos\beta_2+\cos\gamma_1\cos\gamma_2 &= \frac{a_1}{\|\mathbf{a}\|}\frac{b_1}{\|\mathbf{b}\|}+\frac{a_2}{\|\mathbf{a}\|}\frac{b_2}{\|\mathbf{b}\|}+\frac{a_3}{\|\mathbf{a}\|}\frac{b_3}{\|\mathbf{b}\|}\\ &= \frac{1}{\|\mathbf{a}\|\,\|\mathbf{b}\|}(a_1b_1+a_2b_2+a_3b_3) = \frac{1}{\|\mathbf{a}\|\,\|\mathbf{b}\|}(\mathbf{a}\cdot\mathbf{b}) = 0.\end{aligned}$$

31. $\mathbf{a} = \langle 5,7,4\rangle;\ \ \|\mathbf{a}\| = 3\sqrt{10}\,;\ \ \cos\alpha = 5/3\sqrt{10}\,,\ \ \alpha \approx 58.19°;\ \ \cos\beta = 7/3\sqrt{10}\,,\ \ \beta \approx 42.45°;\ \ \cos\gamma = 4/3\sqrt{10}\,,$ $\gamma \approx 65.06°$

32. We want $\cos\alpha = \cos\beta = \cos\gamma$ or $a_1 = a_2 = a_3$. Letting $a_1 = a_2 = a_3 = 1$ we obtain the vector $\mathbf{i}+\mathbf{j}+\mathbf{k}$. A unit vector in the same direction is $\frac{1}{\sqrt{3}}\mathbf{i}+\frac{1}{\sqrt{3}}\mathbf{j}+\frac{1}{\sqrt{3}}\mathbf{k}$.

33. $\text{comp}_{\mathbf{b}}\mathbf{a} = \mathbf{a}\cdot\mathbf{b}/\|\mathbf{b}\| = \langle 1,-1,3\rangle\cdot\langle 2,6,3\rangle/7 = 5/7$

34. $\text{comp}_{\mathbf{a}}\mathbf{b} = \mathbf{b}\cdot\mathbf{a}/\|\mathbf{a}\| = \langle 2,6,3\rangle\cdot\langle 1,-1,3\rangle/\sqrt{11} = 5/\sqrt{11}$

35. $\mathbf{b}-\mathbf{a} = \langle 1,7,0\rangle;\ \ \text{comp}_{\mathbf{a}}(\mathbf{b}-\mathbf{a}) = (\mathbf{b}-\mathbf{a})\cdot\mathbf{a}/\|\mathbf{a}\| = \langle 1,7,0\rangle\cdot\langle 1,-1,3\rangle/\sqrt{11} = -6/\sqrt{11}$

36. $\mathbf{a}+\mathbf{b} = \langle 3,5,6\rangle;\ \ 2\mathbf{b} = \langle 4,12,6\rangle;\ \ \text{comp}_{2\mathbf{b}}(\mathbf{a}+\mathbf{b})\cdot 2\mathbf{b}/|2\mathbf{b}| = \langle 3,5,6\rangle\cdot\langle 4,12,6\rangle/14 = 54/7$

37. $\overrightarrow{OP} = 3\mathbf{i}+10\mathbf{j};\ \ \|\overrightarrow{OP}\| = \sqrt{109}\,;\ \ \text{comp}_{\overrightarrow{OP}}\mathbf{a} = \mathbf{a}\cdot\overrightarrow{OP}/\|\overrightarrow{OP}\| = (4\mathbf{i}+6\mathbf{j})\cdot(3\mathbf{i}+10\mathbf{j})/\sqrt{109} = 72/\sqrt{109}$

38. $\overrightarrow{OP} = \langle 1,-1,1\rangle;\ \ \|\overrightarrow{OP}\| = \sqrt{3}\,;\ \ \text{comp}_{\overrightarrow{OP}}\mathbf{a} = \mathbf{a}\cdot\overrightarrow{OP}/\|\overrightarrow{OP}\| = \langle 2,1,-1\rangle\cdot\langle 1,-1,1\rangle/\sqrt{3} = 0$

39. $\text{comp}_{\mathbf{b}}\mathbf{a} = \mathbf{a}\cdot\mathbf{b}/\|\mathbf{b}\| = (-5\mathbf{i}+5\mathbf{j})\cdot(-3\mathbf{i}+4\mathbf{j})/5 = 7$
$\text{proj}_{\mathbf{b}}\mathbf{a} = (\text{comp}_{\mathbf{b}}\mathbf{a})\mathbf{b}/\|\mathbf{b}\| = 7(-3\mathbf{i}+4\mathbf{j})/5 = -\frac{21}{5}\mathbf{i}+\frac{28}{5}\mathbf{j}$

40. $\text{comp}_{\mathbf{b}}\mathbf{a} = \mathbf{a}\cdot\mathbf{b}/\|\mathbf{b}\| = (4\mathbf{i}+2\mathbf{j})\cdot(-3\mathbf{i}+\mathbf{j})/\sqrt{10} = -\sqrt{10}$
$\text{proj}_{\mathbf{b}}\mathbf{a} = (\text{comp}_{\mathbf{b}}\mathbf{a})\mathbf{b}/\|\mathbf{b}\| = -\sqrt{10}(-3\mathbf{i}+\mathbf{j})/\sqrt{10} = 3\mathbf{i}-\mathbf{j}$

41. $\text{comp}_{\mathbf{b}}\mathbf{a} = \mathbf{a}\cdot\mathbf{b}/\|\mathbf{b}\| = (-\mathbf{i} - 2\mathbf{j} + 7\mathbf{k})\cdot(6\mathbf{i} - 3\mathbf{j} - 2\mathbf{k})/7 = -2$

$\text{proj}_{\mathbf{b}}\mathbf{a} = (\text{comp}_{\mathbf{b}}\mathbf{a})\mathbf{b}/\|\mathbf{b}\| = -2(6\mathbf{i} - 3\mathbf{j} - 2\mathbf{k})/7 = -\frac{12}{7}\mathbf{i} + \frac{6}{7}\mathbf{j} + \frac{4}{7}\mathbf{k}$

42. $\text{comp}_{\mathbf{b}}\mathbf{a} = \mathbf{a}\cdot\mathbf{b}/\|\mathbf{b}\| = \langle 1,1,1\rangle\cdot\langle -2,2,-1\rangle/3 = -1/3$

$\text{proj}_{\mathbf{b}}\mathbf{a} = (\text{comp}_{\mathbf{b}}\mathbf{a})\mathbf{b}/\|\mathbf{b}\| = -\frac{1}{3}\langle -2,2,-1\rangle/3 = \langle 2/9, -2/9, 1/9\rangle$

43. $\mathbf{a}+\mathbf{b} = 3\mathbf{i} + 4\mathbf{j}$; $\|\mathbf{a}+\mathbf{b}\| = 5$; $\text{comp}_{(\mathbf{a}+\mathbf{b})}\mathbf{a} = \mathbf{a}\cdot(\mathbf{a}+\mathbf{b})/\|\mathbf{a}+\mathbf{b}\| = (4\mathbf{i}+3\mathbf{j})\cdot(3\mathbf{i}+4\mathbf{j})/5 = 24/5$

$\text{proj}_{(\mathbf{a}+\mathbf{b})}\mathbf{a} = (\text{comp}_{(\mathbf{a}+\mathbf{b})}\mathbf{a})(\mathbf{a}+\mathbf{b})/\|\mathbf{a}+\mathbf{b}\| = \frac{24}{5}(3\mathbf{i}+4\mathbf{j})/5 = \frac{72}{25}\mathbf{i} + \frac{96}{25}\mathbf{j}$

44. $\mathbf{a}-\mathbf{b} = 5\mathbf{i} + 2\mathbf{j}$; $\|\mathbf{a}-\mathbf{b}\| = \sqrt{29}$; $\text{comp}_{(\mathbf{a}-\mathbf{b})}\mathbf{b} = \mathbf{b}\cdot(\mathbf{a}-\mathbf{b})/\|\mathbf{a}-\mathbf{b}\| = (-\mathbf{i}+\mathbf{j})\cdot(5\mathbf{i}+2\mathbf{j})/\sqrt{29} = -3/\sqrt{29}$

$\text{proj}_{(\mathbf{a}-\mathbf{b})}\mathbf{b} = (\text{comp}_{(\mathbf{a}-\mathbf{b})}\mathbf{b})(\mathbf{a}-\mathbf{b})/\|\mathbf{a}-\mathbf{b}\| = -\frac{3}{\sqrt{29}}(5\mathbf{i}+2\mathbf{j})/\sqrt{29} = -\frac{15}{29}\mathbf{i} - \frac{6}{29}\mathbf{j}$

45. We identify $\|\mathbf{F}\| = 20$, $\theta = 60°$ and $\|\mathbf{d}\| = 100$. Then $W = \|\mathbf{F}\|\,\|\mathbf{d}\|\cos\theta = 20(100)(\frac{1}{2}) = 1000$ ft-lb.

46. We identify $\mathbf{d} = -\mathbf{i} + 3\mathbf{j} + 8\mathbf{k}$. Then $W = \mathbf{F}\cdot\mathbf{d} = \langle 4,3,5\rangle\cdot\langle -1,3,8\rangle = 45$ N-m.

47. **(a)** Since $\mathbf{w}$ and $\mathbf{d}$ are orthogonal, $W = \mathbf{w}\cdot\mathbf{d} = 0$.

(b) We identify $\theta = 0°$. Then $W = \|\mathbf{F}\|\,\|\mathbf{d}\|\cos\theta = 30(\sqrt{4^2+3^2}\,) = 150$ N-m.

48. Using $\mathbf{d} = 6\mathbf{i} + 2\mathbf{j}$ and $\mathbf{F} = 3(\frac{3}{5}\mathbf{i} + \frac{4}{5}\mathbf{j})$, $W = \mathbf{F}\cdot\mathbf{d} = \langle\frac{9}{5}, \frac{12}{5}\rangle\cdot\langle 6,2\rangle = \frac{78}{5}$ ft-lb.

49. Let $\mathbf{a}$ and $\mathbf{b}$ be vectors from the center of the carbon atom to the centers of two distinct hydrogen atoms. The distance between two hydrogen atoms is then

$$\begin{aligned}\|\mathbf{b}-\mathbf{a}\| &= \sqrt{(\mathbf{b}-\mathbf{a})\cdot(\mathbf{b}-\mathbf{a})} = \sqrt{\mathbf{b}\cdot\mathbf{b} - 2\mathbf{a}\cdot\mathbf{b} + \mathbf{a}\cdot\mathbf{a}}\\ &= \sqrt{\|\mathbf{b}\|^2 + \|\mathbf{a}\|^2 - 2\|\mathbf{a}\|\,\|\mathbf{b}\|\cos\theta} = \sqrt{(1.1)^2 + (1.1)^2 - 2(1.1)(1.1)\cos 109.5°}\\ &= \sqrt{1.21 + 1.21 - 2.42(-0.333807)} \approx 1.80 \text{ angstroms.}\end{aligned}$$

50. Using the fact that $|\cos\theta| \le 1$, we have $|\mathbf{a}\cdot\mathbf{b}| = \|\mathbf{a}\|\,\|\mathbf{b}\||\cos\theta| = \|\mathbf{a}\|\,\|\mathbf{b}\||\cos\theta| \le \|\mathbf{a}\|\,\|\mathbf{b}\|$.

51.
$$\begin{aligned}\|\mathbf{a}+\mathbf{b}\|^2 &= (\mathbf{a}+\mathbf{b})\cdot(\mathbf{a}+\mathbf{b}) = \mathbf{a}\cdot\mathbf{a} + 2\mathbf{a}\cdot\mathbf{b} + \mathbf{b}\cdot\mathbf{b} = \|\mathbf{a}\|^2 + 2\mathbf{a}\cdot\mathbf{b} + \|\mathbf{b}\|^2\\ &\le \|\mathbf{a}\|^2 + 2|\mathbf{a}\cdot\mathbf{b}| + \|\mathbf{b}\|^2 \qquad \boxed{\text{since } x \le |x|}\\ &\le \|\mathbf{a}\|^2 + 2\|\mathbf{a}\|\,\|\mathbf{b}\| + \|\mathbf{b}\|^2 = (\|\mathbf{a}\| + \|\mathbf{b}\|)^2 \qquad \boxed{\text{by Problem 50}}\end{aligned}$$

Thus, since $\|\mathbf{a}+\mathbf{b}\|$ and $\|\mathbf{a}\| + \|\mathbf{b}\|$ are positive, $\|\mathbf{a}+\mathbf{b}\| \le \|\mathbf{a}\| + \|\mathbf{b}\|$.

52. Let $P_1(x_1, y_1)$ and $P_2(x_2, y_2)$ be distinct points on the line $ax + by = -c$. Then

$$\begin{aligned}\mathbf{n}\cdot\overrightarrow{P_1P_2} &= \langle a,b\rangle\cdot\langle x_2 - x_1, y_2 - y_1\rangle = ax_2 - ax_1 + by_2 - by_1\\ &= (ax_2 + by_2) - (ax_1 + by_1) = -c - (-c) = 0,\end{aligned}$$

and the vectors are perpendicular. Thus, $\mathbf{n}$ is perpendicular to the line.

53. Let θ be the angle between $\mathbf{n}$ and $\overrightarrow{P_2P_1}$. Then

$$\begin{aligned}d = \|\overrightarrow{P_1P_2}\|\,|\cos\theta| &= \frac{|\mathbf{n}\cdot\overrightarrow{P_2P_1}|}{\|\mathbf{n}\|} = \frac{|\langle a,b\rangle\cdot\langle x_1 - x_2, y_1 - y_2\rangle|}{\sqrt{a^2+b^2}} = \frac{|ax_1 - ax_2 + by_1 - by_2|}{\sqrt{a^2+b^2}}\\ &= \frac{|ax_1 + by_1 - (ax_2 + by_2)|}{\sqrt{a^2+b^2}} = \frac{|ax_1 + by_1 - (-c)|}{\sqrt{a^2+b^2}} = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2+b^2}}.\end{aligned}$$

EXERCISES 7.4

Cross Product

1. $\mathbf{a}\times\mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & -1 & 0 \\ 0 & 3 & 5 \end{vmatrix} = \begin{vmatrix} -1 & 0 \\ 3 & 5 \end{vmatrix}\mathbf{i} - \begin{vmatrix} 1 & 0 \\ 0 & 5 \end{vmatrix}\mathbf{j} + \begin{vmatrix} 1 & -1 \\ 0 & 3 \end{vmatrix}\mathbf{k} = -5\mathbf{i} - 5\mathbf{j} + 3\mathbf{k}$

2. $\mathbf{a}\times\mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 1 & 0 \\ 4 & 0 & -1 \end{vmatrix} = \begin{vmatrix} 1 & 0 \\ 0 & -1 \end{vmatrix}\mathbf{i} - \begin{vmatrix} 2 & 0 \\ 4 & -1 \end{vmatrix}\mathbf{j} + \begin{vmatrix} 2 & 1 \\ 4 & 0 \end{vmatrix}\mathbf{k} = -\mathbf{i} + 2\mathbf{j} - 4\mathbf{k}$

3. $\mathbf{a}\times\mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & -3 & 1 \\ 2 & 0 & 4 \end{vmatrix} = \begin{vmatrix} -3 & 1 \\ 0 & 4 \end{vmatrix}\mathbf{i} - \begin{vmatrix} 1 & 1 \\ 2 & 4 \end{vmatrix}\mathbf{j} + \begin{vmatrix} 1 & -3 \\ 2 & 0 \end{vmatrix}\mathbf{k} = \langle -12, -2, 6\rangle$

4. $\mathbf{a}\times\mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 1 & 1 \\ -5 & 2 & 3 \end{vmatrix} = \begin{vmatrix} 1 & 1 \\ 2 & 3 \end{vmatrix}\mathbf{i} - \begin{vmatrix} 1 & 1 \\ -5 & 3 \end{vmatrix}\mathbf{j} + \begin{vmatrix} 1 & 1 \\ -5 & 2 \end{vmatrix}\mathbf{k} = \langle 1, -8, 7\rangle$

5. $\mathbf{a}\times\mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & -1 & 2 \\ -1 & 3 & -1 \end{vmatrix} = \begin{vmatrix} -1 & 2 \\ 3 & -1 \end{vmatrix}\mathbf{i} - \begin{vmatrix} 2 & 2 \\ -1 & -1 \end{vmatrix}\mathbf{j} + \begin{vmatrix} 2 & -1 \\ -1 & 3 \end{vmatrix}\mathbf{k} = -5\mathbf{i} + 5\mathbf{k}$

6. $\mathbf{a}\times\mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 4 & 1 & -5 \\ 2 & 3 & -1 \end{vmatrix} = \begin{vmatrix} 1 & -5 \\ 3 & -1 \end{vmatrix}\mathbf{i} - \begin{vmatrix} 4 & -5 \\ 2 & -1 \end{vmatrix}\mathbf{j} + \begin{vmatrix} 4 & 1 \\ 2 & 3 \end{vmatrix}\mathbf{k} = 14\mathbf{i} - 6\mathbf{j} + 10\mathbf{k}$

7. $\mathbf{a}\times\mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1/2 & 0 & 1/2 \\ 4 & 6 & 0 \end{vmatrix} = \begin{vmatrix} 0 & 1/2 \\ 6 & 0 \end{vmatrix}\mathbf{i} - \begin{vmatrix} 1/2 & 1/2 \\ 4 & 0 \end{vmatrix}\mathbf{j} + \begin{vmatrix} 1/2 & 0 \\ 4 & 6 \end{vmatrix}\mathbf{k} = \langle -3, 2, 3\rangle$

8. $\mathbf{a}\times\mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 5 & 0 \\ 2 & -3 & 4 \end{vmatrix} = \begin{vmatrix} 5 & 0 \\ -3 & 4 \end{vmatrix}\mathbf{i} - \begin{vmatrix} 0 & 0 \\ 2 & 4 \end{vmatrix}\mathbf{j} + \begin{vmatrix} 0 & 5 \\ 2 & -3 \end{vmatrix}\mathbf{k} = \langle 20, 0, -10\rangle$

9. $\mathbf{a}\times\mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 2 & -4 \\ -3 & -3 & 6 \end{vmatrix} = \begin{vmatrix} 2 & -4 \\ -3 & 6 \end{vmatrix}\mathbf{i} - \begin{vmatrix} 2 & -4 \\ -3 & 6 \end{vmatrix}\mathbf{j} + \begin{vmatrix} 2 & 2 \\ -3 & -3 \end{vmatrix}\mathbf{k} = \langle 0, 0, 0\rangle$

10. $\mathbf{a}\times\mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 8 & 1 & -6 \\ 1 & -2 & 10 \end{vmatrix} = \begin{vmatrix} 1 & -6 \\ -2 & 10 \end{vmatrix}\mathbf{i} - \begin{vmatrix} 8 & -6 \\ 1 & 10 \end{vmatrix}\mathbf{j} + \begin{vmatrix} 8 & 1 \\ 1 & -2 \end{vmatrix}\mathbf{k} = \langle -2, -86, -17\rangle$

11. $\overrightarrow{P_1P_2} = (-2, 2, -4)$; $\overrightarrow{P_1P_3} = (-3, 1, 1)$

$$\overrightarrow{P_1P_2}\times\overrightarrow{P_1P_3} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -2 & 2 & -4 \\ -3 & 1 & 1 \end{vmatrix} = \begin{vmatrix} 2 & -4 \\ 1 & 1 \end{vmatrix}\mathbf{i} - \begin{vmatrix} -2 & -4 \\ -3 & 1 \end{vmatrix}\mathbf{j} + \begin{vmatrix} -2 & 2 \\ -3 & 1 \end{vmatrix}\mathbf{k} = 6\mathbf{i} + 14\mathbf{j} + 4\mathbf{k}$$

12. $\overrightarrow{P_1P_2} = (0,1,1)$; $\overrightarrow{P_1P_3} = (1,2,2)$; $\overrightarrow{P_1P_2} \times \overrightarrow{P_1P_3} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 1 & 1 \\ 1 & 2 & 2 \end{vmatrix} = \begin{vmatrix} 1 & 1 \\ 2 & 2 \end{vmatrix}\mathbf{i} - \begin{vmatrix} 0 & 1 \\ 1 & 2 \end{vmatrix}\mathbf{j} + \begin{vmatrix} 0 & 1 \\ 1 & 2 \end{vmatrix}\mathbf{k} = \mathbf{j} - \mathbf{k}$

13. $\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 7 & -4 \\ 1 & 1 & -1 \end{vmatrix} = \begin{vmatrix} 7 & -4 \\ 1 & -1 \end{vmatrix}\mathbf{i} - \begin{vmatrix} 2 & -4 \\ 1 & -1 \end{vmatrix}\mathbf{j} + \begin{vmatrix} 2 & 7 \\ 1 & 1 \end{vmatrix}\mathbf{k} = -3\mathbf{i} - 2\mathbf{j} - 5\mathbf{k}$

is perpendicular to both $\mathbf{a}$ and $\mathbf{b}$.

14. $\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -1 & -2 & 4 \\ 4 & -1 & 0 \end{vmatrix} = \begin{vmatrix} -2 & 4 \\ -1 & 0 \end{vmatrix}\mathbf{i} - \begin{vmatrix} -1 & 4 \\ 4 & 0 \end{vmatrix}\mathbf{j} + \begin{vmatrix} -1 & -2 \\ 4 & -1 \end{vmatrix}\mathbf{k} = \langle 4, 16, 9 \rangle$

is perpendicular to both $\mathbf{a}$ and $\mathbf{b}$.

15. $\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 5 & -2 & 1 \\ 2 & 0 & -7 \end{vmatrix} = \begin{vmatrix} -2 & 1 \\ 0 & -7 \end{vmatrix}\mathbf{i} - \begin{vmatrix} 5 & 1 \\ 2 & -7 \end{vmatrix}\mathbf{j} + \begin{vmatrix} 5 & -2 \\ 2 & 0 \end{vmatrix}\mathbf{k} = \langle 14, 37, 4 \rangle$

$\mathbf{a} \cdot (\mathbf{a} \times \mathbf{b}) = \langle 5, -2, -1 \rangle \cdot \langle 14, 37, 4 \rangle = 70 - 74 + 4 = 0$; $\mathbf{b} \cdot (\mathbf{a} \times \mathbf{b}) = \langle 2, 0, -7 \rangle \cdot \langle 14, 37, 4 \rangle = 28 + 0 - 28 = 0$

16. $\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1/2 & -1/4 & 0 \\ 2 & -2 & 6 \end{vmatrix} = \begin{vmatrix} -1/4 & 0 \\ -2 & 6 \end{vmatrix}\mathbf{i} - \begin{vmatrix} 1/2 & 0 \\ 2 & 6 \end{vmatrix}\mathbf{j} + \begin{vmatrix} 1/2 & -1/4 \\ 2 & -2 \end{vmatrix}\mathbf{k} = -\frac{3}{2}\mathbf{i} - 3\mathbf{j} - \frac{1}{2}\mathbf{k}$

$\mathbf{a} \cdot (\mathbf{a} \times \mathbf{b}) = (\frac{1}{2}\mathbf{i} - \frac{1}{4}\mathbf{j}) \cdot (-\frac{3}{2}\mathbf{i} - 3\mathbf{j} - \frac{1}{2}\mathbf{k}) = -\frac{3}{4} + \frac{3}{4} + 0 = 0$
$\mathbf{b} \cdot (\mathbf{a} \times \mathbf{b}) = (2\mathbf{i} - 2\mathbf{j} + 6\mathbf{k}) \cdot (-\frac{3}{2}\mathbf{i} - 3\mathbf{j} - \frac{1}{2}\mathbf{k}) = -3 + 6 - 3 = 0$

17. **(a)** $\mathbf{b} \times \mathbf{c} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 1 & 1 \\ 3 & 1 & 1 \end{vmatrix} = \begin{vmatrix} 1 & 1 \\ 1 & 1 \end{vmatrix}\mathbf{i} - \begin{vmatrix} 2 & 1 \\ 3 & 1 \end{vmatrix}\mathbf{j} + \begin{vmatrix} 2 & 1 \\ 3 & 1 \end{vmatrix}\mathbf{k} = \mathbf{j} - \mathbf{k}$

$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & -1 & 2 \\ 0 & 1 & -1 \end{vmatrix} = \begin{vmatrix} -1 & 2 \\ 1 & -1 \end{vmatrix}\mathbf{i} - \begin{vmatrix} 1 & 2 \\ 0 & -1 \end{vmatrix}\mathbf{j} + \begin{vmatrix} 1 & -1 \\ 0 & 1 \end{vmatrix}\mathbf{k} = -\mathbf{i} + \mathbf{j} + \mathbf{k}$

(b) $\mathbf{a} \cdot \mathbf{c} = (\mathbf{i} - \mathbf{j} + 2\mathbf{k}) \cdot (3\mathbf{i} + \mathbf{j} + \mathbf{k}) = 4$; $(\mathbf{a} \cdot \mathbf{c})\mathbf{b} = 4(2\mathbf{i} + \mathbf{j} + \mathbf{k}) = 8\mathbf{i} + 4\mathbf{j} + 4\mathbf{k}$
$\mathbf{a} \cdot \mathbf{b} = (\mathbf{i} - \mathbf{j} + 2\mathbf{k}) \cdot (2\mathbf{i} + \mathbf{j} + \mathbf{k}) = 3$; $(\mathbf{a} \cdot \mathbf{b})\mathbf{c} = 3(3\mathbf{i} + \mathbf{j} + \mathbf{k}) = 9\mathbf{i} + 3\mathbf{j} + 3\mathbf{k}$
$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{a} \cdot \mathbf{b})\mathbf{c} = (8\mathbf{i} + 4\mathbf{j} + 4\mathbf{k}) - (9\mathbf{i} + 3\mathbf{j} + 3\mathbf{k}) = -\mathbf{i} + \mathbf{j} + \mathbf{k}$

18. **(a)** $\mathbf{b} \times \mathbf{c} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 2 & -1 \\ -1 & 5 & 8 \end{vmatrix} = \begin{vmatrix} 2 & -1 \\ 5 & 8 \end{vmatrix}\mathbf{i} - \begin{vmatrix} 1 & -1 \\ -1 & 8 \end{vmatrix}\mathbf{j} + \begin{vmatrix} 1 & 2 \\ -1 & 5 \end{vmatrix}\mathbf{k} = 21\mathbf{i} - 7\mathbf{j} + 7\mathbf{k}$

$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3 & 0 & -4 \\ 21 & -7 & 7 \end{vmatrix} = \begin{vmatrix} 0 & -4 \\ -7 & 7 \end{vmatrix}\mathbf{i} - \begin{vmatrix} 3 & -4 \\ 21 & 7 \end{vmatrix}\mathbf{j} + \begin{vmatrix} 3 & 0 \\ 21 & -7 \end{vmatrix}\mathbf{k} = -28\mathbf{i} - 105\mathbf{j} - 21\mathbf{k}$

(b) $\mathbf{a} \cdot \mathbf{c} = (3\mathbf{i} - 4\mathbf{k}) \cdot (-\mathbf{i} + 5\mathbf{j} + 8\mathbf{k}) = -35$; $(\mathbf{a} \cdot \mathbf{c})\mathbf{b} = -35(\mathbf{i} + 2\mathbf{j} - \mathbf{k}) = -35\mathbf{i} - 70\mathbf{j} + 35\mathbf{k}$
$\mathbf{a} \cdot \mathbf{b} = (3\mathbf{i} - 4\mathbf{k}) \cdot (\mathbf{i} + 2\mathbf{j} - \mathbf{k}) = 7$; $(\mathbf{a} \cdot \mathbf{b})\mathbf{c} = 7(-\mathbf{i} + 5\mathbf{j} + 8\mathbf{k}) = -7\mathbf{i} + 35\mathbf{j} + 56\mathbf{k}$
$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{a} \cdot \mathbf{b})\mathbf{c} = (-35\mathbf{i} - 70\mathbf{j} + 35\mathbf{k}) - (-7\mathbf{i} + 35\mathbf{j} + 56\mathbf{k}) = -28\mathbf{i} - 105\mathbf{j} - 21\mathbf{k}$

19. $(2\mathbf{i}) \times \mathbf{j} = 2(\mathbf{i} \times \mathbf{j}) = 2\mathbf{k}$

20. $\mathbf{i} \times (-3\mathbf{k}) = -3(\mathbf{i} \times \mathbf{k}) = -3(-\mathbf{j}) = 3\mathbf{j}$

21. $\mathbf{k}\times(2\mathbf{i}-\mathbf{j})=\mathbf{k}\times(2\mathbf{i})+\mathbf{k}\times(-\mathbf{j})=2(\mathbf{k}\times\mathbf{i})-(\mathbf{k}\times\mathbf{j})=2\mathbf{j}-(-\mathbf{i})=\mathbf{i}+2\mathbf{j}$

22. $\mathbf{i}\times(\mathbf{j}\times\mathbf{k})=\mathbf{i}\times\mathbf{i}=\mathbf{0}$

23. $[(2\mathbf{k})\times(3\mathbf{j})]\times(4\mathbf{j})=[2\cdot 3(\mathbf{k}\times\mathbf{j})\times(4\mathbf{j})]=6(-\mathbf{i})\times 4\mathbf{j}=(-6)(4)(\mathbf{i}\times\mathbf{j})=-24\mathbf{k}$

24. $(2\mathbf{i}-\mathbf{j}+5\mathbf{k})\times\mathbf{i}=(2\mathbf{i}\times\mathbf{i})+(-\mathbf{j}\times\mathbf{i})+(5\mathbf{k}\times\mathbf{i})=2(\mathbf{i}\times\mathbf{i})+(\mathbf{i}\times\mathbf{j})+5(\mathbf{k}\times\mathbf{i})=5\mathbf{j}+\mathbf{k}$

25.
$$\begin{aligned}(\mathbf{i}+\mathbf{j})\times(\mathbf{i}+5\mathbf{k})&=[(\mathbf{i}+\mathbf{j})\times\mathbf{i}]+[(\mathbf{i}+\mathbf{j})\times 5\mathbf{k}]=(\mathbf{i}\times\mathbf{i})+(\mathbf{j}\times\mathbf{i})+(\mathbf{i}\times 5\mathbf{k})+(\mathbf{j}\times 5\mathbf{k})\\&=-\mathbf{k}+5(-\mathbf{j})+5\mathbf{i}=5\mathbf{i}-5\mathbf{j}-\mathbf{k}\end{aligned}$$

26. $\mathbf{i}\times\mathbf{k}-2(\mathbf{j}\times\mathbf{i})=-\mathbf{j}-2(-\mathbf{k})=-\mathbf{j}+2\mathbf{k}$

27. $\mathbf{k}\cdot(\mathbf{j}\times\mathbf{k})=\mathbf{k}\cdot\mathbf{i}=0$

28. $\mathbf{i}\cdot[\mathbf{j}\times(-\mathbf{k})]=\mathbf{i}\cdot[-(\mathbf{j}\times\mathbf{k})]=\mathbf{i}\cdot(-\mathbf{i})=-(\mathbf{i}\cdot\mathbf{i})=-1$

29. $\|4\mathbf{j}-5(\mathbf{i}\times\mathbf{j})\|=\|4\mathbf{j}-5\mathbf{k}\|=\sqrt{41}$

30. $(\mathbf{i}\times\mathbf{j})\cdot(3\mathbf{j}\times\mathbf{i})=\mathbf{k}\cdot(-3\mathbf{k})=-3(\mathbf{k}\cdot\mathbf{k})=-3$

31. $\mathbf{i}\times(\mathbf{i}\times\mathbf{j})=\mathbf{i}\times\mathbf{k}=-\mathbf{j}$

32. $(\mathbf{i}\times\mathbf{j})\times\mathbf{i}=\mathbf{k}\times\mathbf{i}=\mathbf{j}$

33. $(\mathbf{i}\times\mathbf{i})\times\mathbf{j}=\mathbf{0}\times\mathbf{j}=\mathbf{0}$

34. $(\mathbf{i}\cdot\mathbf{i})(\mathbf{i}\times\mathbf{j})=1(\mathbf{k})=\mathbf{k}$

35.
$$\begin{aligned}2\mathbf{j}\cdot[\mathbf{i}\times(\mathbf{j}-3\mathbf{k})]&=2\mathbf{j}\cdot[(\mathbf{i}\times\mathbf{j})+(\mathbf{i}\times(-3\mathbf{k})]=2\mathbf{j}\cdot[\mathbf{k}+3(\mathbf{k}\times\mathbf{i})]=2\mathbf{j}\cdot(\mathbf{k}+3\mathbf{j})=2\mathbf{j}\cdot\mathbf{k}+2\mathbf{j}\cdot 3\mathbf{j}\\&=2(\mathbf{j}\cdot\mathbf{k})+6(\mathbf{j}\cdot\mathbf{j})=2(0)+6(1)=6\end{aligned}$$

36. $(\mathbf{i}\times\mathbf{k})\times(\mathbf{j}\times\mathbf{i})=(-\mathbf{j})\times(-\mathbf{k})=(-1)(-1)(\mathbf{j}\times\mathbf{k})=\mathbf{j}\times\mathbf{k}=\mathbf{i}$

37. $\mathbf{a}\times(3\mathbf{b})=3(\mathbf{a}\times\mathbf{b})=3(4\mathbf{i}-3\mathbf{j}+6\mathbf{k})=12\mathbf{i}-9\mathbf{j}+18\mathbf{k}$

38. $\mathbf{b}\times\mathbf{a}=-\mathbf{a}\times\mathbf{b}=-(\mathbf{a}\times\mathbf{b})=-4\mathbf{i}+3\mathbf{j}-6\mathbf{k}$

39. $(-\mathbf{a})\times\mathbf{b}=-(\mathbf{a}\times\mathbf{b})=-4\mathbf{i}+3\mathbf{j}-6\mathbf{k}$

40. $|\mathbf{a}\times\mathbf{b}|=\sqrt{4^2+(-3)^2+6^2}=\sqrt{61}$

41. $(\mathbf{a}\times\mathbf{b})\times\mathbf{c}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\4&-3&6\\2&4&-1\end{vmatrix}=\begin{vmatrix}-3&6\\4&-1\end{vmatrix}\mathbf{i}-\begin{vmatrix}4&6\\2&-1\end{vmatrix}\mathbf{j}+\begin{vmatrix}4&-3\\2&-4\end{vmatrix}\mathbf{k}=-21\mathbf{i}+16\mathbf{j}+22\mathbf{k}$

42. $(\mathbf{a}\times\mathbf{b})\cdot\mathbf{c}=4(2)+(-3)4+6(-1)=-10$

43. $\mathbf{a}\cdot(\mathbf{b}\times\mathbf{c})=(\mathbf{a}\times\mathbf{b})\cdot\mathbf{c}=4(2)+(-3)4+6(-1)=-10$

44. $(4\mathbf{a})\cdot(\mathbf{b}\times\mathbf{c})=(4\mathbf{a}\times\mathbf{b})\cdot\mathbf{c}=4(\mathbf{a}\times\mathbf{b})\cdot\mathbf{c}=16(2)+(-12)4+24(-1)=-40$

45. (a) Let $A=(1,3,0)$, $B=(2,0,0)$, $C=(0,0,4)$, and $D=(1,-3,4)$. Then $\overrightarrow{AB}=\mathbf{i}-3\mathbf{j}$, $\overrightarrow{AC}=-\mathbf{i}-3\mathbf{j}+4\mathbf{k}$, $\overrightarrow{CD}=\mathbf{i}-3\mathbf{j}$, and $\overrightarrow{BD}=-\mathbf{i}-3\mathbf{j}+4\mathbf{k}$. Since $\overrightarrow{AB}=\overrightarrow{CD}$ and $\overrightarrow{AC}=\overrightarrow{BD}$, the quadrilateral is a parallelogram.

(b) Computing
$$\overrightarrow{AB}\times\overrightarrow{AC}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\1&-3&0\\-1&-3&4\end{vmatrix}=-12\mathbf{i}-4\mathbf{j}-6\mathbf{k}$$
we find that the area is $\|-12\mathbf{i}-4\mathbf{j}-6\mathbf{k}\|=\sqrt{144+16+36}=14$.

46. (a) Let $A=(3,4,1)$, $B=(-1,4,2)$, $C=(2,0,2)$ and $D=(-2,0,3)$. Then $\overrightarrow{AB}=-4\mathbf{i}+\mathbf{k}$, $\overrightarrow{AC}=-\mathbf{i}-4\mathbf{j}+\mathbf{k}$, $\overrightarrow{CD}=-4\mathbf{i}+\mathbf{k}$, and $\overrightarrow{BD}=-\mathbf{i}-4\mathbf{j}+\mathbf{k}$. Since $\overrightarrow{AB}=\overrightarrow{CD}$ and $\overrightarrow{AC}=\overrightarrow{BD}$, the quadrilateral is a parallelogram.

(b) Computing
$$\overrightarrow{AB}\times\overrightarrow{AC}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\-4&0&1\\-1&-4&1\end{vmatrix}=4\mathbf{i}+3\mathbf{j}+16\mathbf{k}$$

we find that the area is $\|4\mathbf{i}+3\mathbf{j}+16\mathbf{k}\| = \sqrt{16+9+256} = \sqrt{281} \approx 16.76$.

47. $\overrightarrow{P_1P_2} = \mathbf{j}$; $\overrightarrow{P_2P_3} = -\mathbf{j}+\mathbf{k}$

$$\overrightarrow{P_1P_2} \times \overrightarrow{P_2P_3} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{vmatrix} = \begin{vmatrix} 1 & 0 \\ -1 & 1 \end{vmatrix}\mathbf{i} - \begin{vmatrix} 0 & 0 \\ 0 & 1 \end{vmatrix}\mathbf{j} + \begin{vmatrix} 0 & 1 \\ 0 & -1 \end{vmatrix}\mathbf{k} = \mathbf{i}; \quad A = \tfrac{1}{2}\|\mathbf{i}\| = \tfrac{1}{2} \text{ sq. unit}$$

48. $\overrightarrow{P_1P_2} = \mathbf{j}+2\mathbf{k}$; $\overrightarrow{P_2P_3} = 2\mathbf{i}+\mathbf{j}-2\mathbf{k}$

$$\overrightarrow{P_1P_2} \times \overrightarrow{P_2P_3} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 1 & 2 \\ 2 & 1 & -2 \end{vmatrix} = \begin{vmatrix} 1 & 2 \\ 1 & -2 \end{vmatrix}\mathbf{i} - \begin{vmatrix} 0 & 2 \\ 2 & -2 \end{vmatrix}\mathbf{j} + \begin{vmatrix} 0 & 1 \\ 2 & 1 \end{vmatrix}\mathbf{k} = -4\mathbf{i}+4\mathbf{j}-2\mathbf{k}$$

$A = \frac{1}{2}\|-4\mathbf{i}+4\mathbf{j}-2\mathbf{k}\| = 3$ sq. units

49. $\overrightarrow{P_1P_2} = -3\mathbf{j}-\mathbf{k}$; $\overrightarrow{P_2P_3} = -2\mathbf{i}-\mathbf{k}$

$$\overrightarrow{P_1P_2} \times \overrightarrow{P_2P_3} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & -3 & -1 \\ -2 & 0 & -1 \end{vmatrix} = \begin{vmatrix} -3 & -1 \\ 0 & -1 \end{vmatrix}\mathbf{i} - \begin{vmatrix} 0 & -1 \\ -2 & -1 \end{vmatrix}\mathbf{j} + \begin{vmatrix} 0 & -3 \\ -2 & 0 \end{vmatrix}\mathbf{k} = 3\mathbf{i}+2\mathbf{j}-6\mathbf{k}$$

$A = \frac{1}{2}\|3\mathbf{i}+2\mathbf{j}-6\mathbf{k}\| = \frac{7}{2}$ sq. units

50. $\overrightarrow{P_1P_2} = -\mathbf{i}+3\mathbf{k}$; $\overrightarrow{P_2P_3} = 2\mathbf{i}+4\mathbf{j}-\mathbf{k}$

$$\overrightarrow{P_1P_2} \times \overrightarrow{P_2P_3} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -1 & 0 & 3 \\ 2 & 4 & -1 \end{vmatrix} = \begin{vmatrix} 0 & 3 \\ 4 & -1 \end{vmatrix}\mathbf{i} - \begin{vmatrix} -1 & 3 \\ 2 & -1 \end{vmatrix}\mathbf{j} + \begin{vmatrix} -1 & 0 \\ 2 & 4 \end{vmatrix}\mathbf{k} = -12\mathbf{i}+5\mathbf{j}-4\mathbf{k}$$

$A = \frac{1}{2}\|-12\mathbf{i}+5\mathbf{j}-4\mathbf{k}\| = \frac{\sqrt{185}}{2}$ sq. units

51. $$\mathbf{b}\times\mathbf{c} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -1 & 4 & 0 \\ 2 & 2 & 2 \end{vmatrix} = \begin{vmatrix} 4 & 0 \\ 2 & 2 \end{vmatrix}\mathbf{i} - \begin{vmatrix} -1 & 0 \\ 2 & 2 \end{vmatrix}\mathbf{j} + \begin{vmatrix} -1 & 4 \\ 2 & 2 \end{vmatrix}\mathbf{k} = 8\mathbf{i}+2\mathbf{j}-10\mathbf{k}$$

$\mathbf{v} = |\mathbf{a}\cdot(\mathbf{b}\times\mathbf{c})| = |(\mathbf{i}+\mathbf{j})\cdot(8\mathbf{i}+2\mathbf{j}-10\mathbf{k})| = |8+2+0| = 10$ cu. units

52. $$\mathbf{b}\times\mathbf{c} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 4 & 1 \\ 1 & 1 & 5 \end{vmatrix} = \begin{vmatrix} 4 & 1 \\ 1 & 5 \end{vmatrix}\mathbf{i} - \begin{vmatrix} 1 & 1 \\ 1 & 5 \end{vmatrix}\mathbf{j} + \begin{vmatrix} 1 & 4 \\ 1 & 1 \end{vmatrix}\mathbf{k} = 19\mathbf{i}-4\mathbf{j}-3\mathbf{k}$$

$\mathbf{v} = |\mathbf{a}\cdot(\mathbf{b}\times\mathbf{c})| = |(3\mathbf{i}+\mathbf{j}+\mathbf{k})\cdot(19\mathbf{i}-4\mathbf{j}-3\mathbf{k})| = |57-4-3| = 50$ cu. units

53. $$\mathbf{b}\times\mathbf{c} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -2 & 6 & -6 \\ 5/2 & 3 & 1/2 \end{vmatrix} = \begin{vmatrix} 6 & -6 \\ 3 & 2/2 \end{vmatrix}\mathbf{i} - \begin{vmatrix} -2 & -6 \\ 5/2 & 1/2 \end{vmatrix}\mathbf{j} + \begin{vmatrix} -2 & 6 \\ 5/2 & 3 \end{vmatrix}\mathbf{k} = 21\mathbf{i}-14\mathbf{j}-21\mathbf{k}$$

$\mathbf{a}\cdot(\mathbf{b}\times\mathbf{c}) = (4\mathbf{i}+6\mathbf{j})\cdot(21\mathbf{i}-14\mathbf{j}-21\mathbf{k}) = 84-84+0 = 0$. The vectors are coplanar.

54. The four points will be coplanar if the three vectors $\overrightarrow{P_1P_2} = \langle 3,-1,-1\rangle$, $\overrightarrow{P_2P_3} = \langle -3,-5,13\rangle$, and $\overrightarrow{P_3P_4} = \langle -8,7,-6\rangle$ are coplanar.

$$\overrightarrow{P_2P_3} \times \overrightarrow{P_3P_4} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -3 & -5 & 13 \\ -8 & 7 & -6 \end{vmatrix} = \begin{vmatrix} -5 & 13 \\ 7 & -6 \end{vmatrix}\mathbf{i} - \begin{vmatrix} -3 & 13 \\ -8 & -6 \end{vmatrix}\mathbf{j} + \begin{vmatrix} -3 & -5 \\ -8 & 7 \end{vmatrix}\mathbf{k} = \langle -61,-122,-61\rangle$$

$\overrightarrow{P_1P_2}\cdot(\overrightarrow{P_2P_3}\times\overrightarrow{P_3P_4}) = \langle 3,-1,-1\rangle\cdot\langle -61,-122,-61\rangle = -183+122+61 = 0$

The four points are coplanar.

55. **(a)** Since $\theta = 90^\circ$, $\|\mathbf{a}\times\mathbf{b}\| = \|\mathbf{a}\|\,\|\mathbf{b}\|\,|\sin 90^\circ| = 6.4(5) = 32$.

(b) The direction of $\mathbf{a}\times\mathbf{b}$ is into the fourth quadrant of the xy-plane or to the left of the plane determined by $\mathbf{a}$ and $\mathbf{b}$ as shown in Figure 7.54 in the text. It makes an angle of 30° with the positive x-axis.

(c) We identify $\mathbf{n}=(\sqrt{3}\,\mathbf{i}-\mathbf{j})/2$. Then $\mathbf{a}\times\mathbf{b}=32\mathbf{n}=16\sqrt{3}\,\mathbf{i}-16\mathbf{j}$.

56. Using Definition 7.4, $\mathbf{a}\times\mathbf{b}=\sqrt{27}\,(8)\sin 120°\mathbf{n}=24\sqrt{3}\,(\sqrt{3}/2)\mathbf{n}=36\mathbf{n}$. By the right-hand rule, $\mathbf{n}=\mathbf{j}$ or $\mathbf{n}=-\mathbf{j}$. Thus, $\mathbf{a}\times\mathbf{b}=36\mathbf{j}$ or $-36\mathbf{j}$.

57. (a) We note first that $\mathbf{a}\times\mathbf{b}=\mathbf{k}$, $\mathbf{b}\times\mathbf{c}=\frac{1}{2}(\mathbf{i}-\mathbf{k})$, $\mathbf{c}\times\mathbf{a}=\frac{1}{2}(\mathbf{j}-\mathbf{k})$, $\mathbf{a}\cdot(\mathbf{b}\times\mathbf{c})=\frac{1}{2}$, $\mathbf{b}\cdot(\mathbf{c}\times\mathbf{a})=\frac{1}{2}$, and $\mathbf{c}\cdot(\mathbf{a}\times\mathbf{b})=\frac{1}{2}$. Then

$$\mathbf{A}=\frac{\frac{1}{2}(\mathbf{i}-\mathbf{k})}{\frac{1}{2}}=\mathbf{i}-\mathbf{k},\quad \mathbf{B}=\frac{\frac{1}{2}(\mathbf{j}-\mathbf{k})}{\frac{1}{2}}=\mathbf{j}-\mathbf{k},\quad\text{and}\quad \mathbf{C}=\frac{\mathbf{k}}{\frac{1}{2}}=2\mathbf{k}.$$

(b) We need to compute $\mathbf{A}\cdot(\mathbf{B}\times\mathbf{C})$. Using formula (10) in the text we have

$$\begin{aligned}\mathbf{B}\times\mathbf{C}&=\frac{(\mathbf{c}\times\mathbf{a})\times(\mathbf{a}\times\mathbf{b})}{[\mathbf{b}\cdot(\mathbf{c}\times\mathbf{a})][\mathbf{c}\cdot(\mathbf{a}\times\mathbf{b})]}=\frac{[(\mathbf{c}\times\mathbf{a})\cdot\mathbf{b}]\mathbf{a}-[(\mathbf{c}\times\mathbf{a})\cdot\mathbf{a}]\mathbf{b}}{[\mathbf{b}\cdot(\mathbf{c}\times\mathbf{a})][\mathbf{c}\cdot(\mathbf{a}\times\mathbf{b})]}\\&=\frac{\mathbf{a}}{\mathbf{c}\cdot(\mathbf{a}\times\mathbf{b})}\qquad\boxed{\text{since }(\mathbf{c}\times\mathbf{a})\cdot\mathbf{a}=0.}\end{aligned}$$

Then

$$\mathbf{A}\cdot(\mathbf{B}\times\mathbf{C})=\frac{\mathbf{b}\times\mathbf{c}}{\mathbf{a}\cdot(\mathbf{b}\times\mathbf{c})}\cdot\frac{\mathbf{a}}{\mathbf{c}\cdot(\mathbf{a}\times\mathbf{b})}=\frac{1}{\mathbf{c}\cdot(\mathbf{a}\times\mathbf{b})}$$

and the volume of the unit cell of the reciprocal latrice is the reciprocal of the volume of the unit cell of the original lattice.

58.
$$\begin{aligned}\mathbf{a}\times(\mathbf{b}+\mathbf{c})&=\begin{vmatrix}a_2 & a_3\\ b_2+c_2 & b_3+c_3\end{vmatrix}\mathbf{i}-\begin{vmatrix}a_1 & a_3\\ b_1+c_1 & b_3+c_3\end{vmatrix}\mathbf{j}+\begin{vmatrix}a_1 & a_2\\ b_1+c_1 & b_2+c_2\end{vmatrix}\mathbf{k}\\&=(a_2b_3-a_3b_2)\mathbf{i}+(a_2c_3-a_3c_2)\mathbf{i}-[(a_1b_3-a_3b_1)\mathbf{j}+(a_1c_3-a_3c_1)\mathbf{j}]+(a_1b_2-a_2b_1)\mathbf{k}+(a_1c_2-a_2c_1)\mathbf{k}\\&=(a_2b_3-a_3b_2)\mathbf{i}-(a_1b_3-a_3b_1)\mathbf{j}+(a_1b_2-a_2b_1)\mathbf{k}+(a_2c_3-a_3c_2)\mathbf{i}-(a_1c_3-a_3c_1)\mathbf{j}+(a_1c_2-a_2c_1)\mathbf{k}\\&=\mathbf{a}\times\mathbf{b}+\mathbf{a}\times\mathbf{c}\end{aligned}$$

59. $\mathbf{b}\times\mathbf{c}=(b_2c_3-b_3c_2)\mathbf{i}-(b_1c_3-b_3c_1)\mathbf{j}+(b_1c_2-b_2c_1)\mathbf{k}$

$$\begin{aligned}\mathbf{a}\times(\mathbf{b}\times\mathbf{c})&=[a_2(b_1c_2-b_2c_1)+a_3(b_1c_3-b_3c_1)]\mathbf{i}-[a_1(b_1c_2-b_2c_1)-a_3(b_2c_3-b_3c_2)]\mathbf{j}\\&\quad+[-a_1(b_1c_3-b_3c_1)-a_2(b_2c_3-b_3c_2)]\mathbf{k}\\&=(a_2b_1c_2-a_2b_2c_1+a_3b_1c_3-a_3b_3c_1)\mathbf{i}-(a_1b_1c_2-a_1b_2c_1-a_3b_2c_3+a_3b_3c_2)\mathbf{j}\\&\quad-(a_1b_1c_3-a_1b_3c_1+a_2b_2c_3-a_2b_3c_2)\mathbf{k}\end{aligned}$$

$$\begin{aligned}(\mathbf{a}\cdot\mathbf{c})\mathbf{b}-(\mathbf{a}\cdot\mathbf{b})\mathbf{c}&=(a_1c_1+a_2c_2+a_3c_3)(b_1\mathbf{i}+b_2\mathbf{j}+b_3\mathbf{k})-(a_1b_1+a_2b_2+a_3b_3)(c_1\mathbf{i}+c_2\mathbf{j}+c_3\mathbf{k})\\&=(a_2b_1c_2-a_2b_2c_1+a_3b_1c_3-a_3b_3c_1)\mathbf{i}-(a_1b_1c_2-a_1b_2c_1-a_3b_2c_3+a_3b_3c_2)\mathbf{j}\\&\quad-(a_1b_1c_3-a_1b_3c_1+a_2b_2c_3-a_2b_3c_2)\mathbf{k}\end{aligned}$$

60. The statement is false since $\mathbf{i}\times(\mathbf{i}\times\mathbf{j})=\mathbf{i}\times\mathbf{k}=-\mathbf{j}$ and $(\mathbf{i}\times\mathbf{i})\times\mathbf{j}=\mathbf{0}\times\mathbf{j}=\mathbf{0}$.

61. Using equation 9 in the text,

$$\mathbf{a}\cdot(\mathbf{b}\times\mathbf{c})=\begin{vmatrix}a_1&a_2&a_3\\b_1&b_2&b_3\\c_1&c_2&c_3\end{vmatrix}\quad\text{and}\quad(\mathbf{a}\times\mathbf{b})\cdot\mathbf{c}=\mathbf{c}\cdot(\mathbf{a}\times\mathbf{b})=\begin{vmatrix}c_1&c_2&c_3\\a_1&a_2&a_3\\b_1&b_2&b_3\end{vmatrix}.$$

Expanding these determinants out we obtain $\mathbf{a}\cdot(\mathbf{b}\times\mathbf{c})=a_1b_2c_3+a_2b_3c_1+a_3b_1c_2-a_3b_2c_1-a_1b_3c_2-a_2b_1c_3$ and $\mathbf{c}\cdot(\mathbf{a}\times\mathbf{b})=a_2b_3c_1+a_3b_1c_2+a_1b_2c_3-a_2b_1c_3-a_3b_2c_1-a_1b_3c_2$. These are equal so $\mathbf{a}\cdot(\mathbf{b}\times\mathbf{c})=(\mathbf{a}\times\mathbf{b})\cdot\mathbf{c}$.

62. $\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) + \mathbf{b} \times (\mathbf{c} \times \mathbf{a}) + \mathbf{c} \times (\mathbf{a} \times \mathbf{b})$

$$= (\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{a} \cdot \mathbf{b})\mathbf{c} + (\mathbf{b} \cdot \mathbf{a})\mathbf{c} - (\mathbf{b} \cdot \mathbf{c})\mathbf{a} + (\mathbf{c} \cdot \mathbf{b})\mathbf{a} - (\mathbf{c} \cdot \mathbf{a})\mathbf{b}$$
$$= [(\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{c} \cdot \mathbf{a})\mathbf{b}] + [(\mathbf{b} \cdot \mathbf{a})\mathbf{c} - (\mathbf{a} \cdot \mathbf{b})\mathbf{c}] + [(\mathbf{c} \cdot \mathbf{b})\mathbf{a} - (\mathbf{b} \cdot \mathbf{c})\mathbf{a}] = \mathbf{0}$$

63. Since

$$\begin{aligned}\|\mathbf{a} \times \mathbf{b}\|^2 &= (a_2b_3 - a_3b_2)^2 + (a_1b_3 - a_3b_1)^2 + (a_1b_2 - a_2b_1)^2 \\ &= a_2^2b_3^2 - 2a_2b_3a_3b_2 + a_3^2b_2^2 + a_1^2b_3^2 - 2a_1b_3a_3b_1 + a_3^2b_1^2 + a_1^2b_2^2 - 2a_1b_2a_2b_1 + a_2^2b_1^2\end{aligned}$$

and

$$\begin{aligned}\|\mathbf{a}\|^2\|\mathbf{b}\|^2 - (\mathbf{a} \cdot \mathbf{b})^2 &= (a_1^2 + a_2^2 + a_3^2)(b_1^2 + b_2^2 + b_3^2) - (a_1b_1 + a_2b_2 + a_3b_3)^2 \\ &= a_1^2a_2^2 + a_1^2b_2^2 + a_1^2b_3^2 + a_2^2b_1^2 + a_2^2b_2^2 + a_2^2b_3^2 + a_3^2b_1^2 + a_3^2b_2^2 + a_3^2b_3^2 \\ &\quad - a_1^2b_1^2 - a_2^2b_2^2 - a_3^2b_3^2 - 2a_1b_1a_2b_2 - 2a_1b_1a_3b_3 - 2a_2b_2a_3b_3 \\ &= a_1^2b_2^2 + a_1^2b_3^2 + a_2^2b_1^2 + a_2^2b_3^2 + a_3^2b_1^2 + a_3^2b_2^2 - 2a_1a_2b_1b_2 - 2a_1a_3b_1b_3 - 2a_2a_3b_2b_3\end{aligned}$$

we see that $\|\mathbf{a} \times \mathbf{b}\|^2 = \|\mathbf{a}\|^2\|\mathbf{b}\|^2 - (\mathbf{a} \cdot \mathbf{b})^2$.

64. No. For example $\mathbf{i} \times (\mathbf{i} + \mathbf{j}) = \mathbf{i} \times \mathbf{j}$ by the distributive law (**iii**) in the text, and the fact that $\mathbf{i} \times \mathbf{i} = \mathbf{0}$. But $\mathbf{i} + \mathbf{j}$ does not equal $\mathbf{j}$.

65. By the distributive law (**iii**) in the text:

$$(\mathbf{a} + \mathbf{b}) \times (\mathbf{a} - \mathbf{b}) = (\mathbf{a} + \mathbf{b}) \times \mathbf{a} - (\mathbf{a} + \mathbf{b}) \times \mathbf{b} = \mathbf{a} \times \mathbf{a} + \mathbf{b} \times \mathbf{a} - \mathbf{a} \times \mathbf{b} - \mathbf{b} \times \mathbf{b} = 2\mathbf{b} \times \mathbf{a}$$

since $\mathbf{a} \times \mathbf{a} = \mathbf{0}$, $\mathbf{b} \times \mathbf{b} = \mathbf{0}$, and $-\mathbf{a} \times \mathbf{b} = \mathbf{b} \times \mathbf{a}$.

EXERCISES 7.5

Lines and Planes in 3-Space

The equation of a line through P_1 and P_2 in 3-space with $\mathbf{r}_1 = \overrightarrow{OP_1}$ and $\mathbf{r}_2 = \overrightarrow{OP_2}$ can be expressed as $\mathbf{r} = \mathbf{r}_1 + t(k\mathbf{a})$ or $\mathbf{r} = \mathbf{r}_2 + t(k\mathbf{a})$ where $\mathbf{a} = \mathbf{r}_2 - \mathbf{r}_1$ and k is any non-zero scalar. Thus, the form of the equation of a line is not unique. (See the alternate solution to Problem 1.)

1. $\mathbf{a} = \langle 1-3, 2-5, 1-(-2)\rangle = \langle -2, -3, 3\rangle$; $\langle x, y, z\rangle = \langle 1, 2, 1\rangle + t\langle -2, -3, 3\rangle$
Alternate Solution: $\mathbf{a} = \langle 3-1, 5-2, -2-1\rangle = \langle 2, 3, -3\rangle$; $\langle x, y, z\rangle = \langle 3, 5, -2\rangle + t\langle 2, 3, -3\rangle$

2. $\mathbf{a} = \langle 0-(-2), 4-6, 5-3\rangle = \langle 2, -2, 2\rangle$; $\langle x, y, z\rangle = \langle 0, 4, 5\rangle + t\langle 2, -2, 2\rangle$

3. $\mathbf{a} = \langle 1/2-(-3/2), -1/2-5/2, 1-(-1/2)\rangle = \langle 2, -3, 3/2\rangle$; $\langle x, y, z\rangle = \langle 1/2, -1/2, 1\rangle + t\langle 2, -3, 3/2\rangle$

4. $\mathbf{a} = \langle 10-5, 2-(-3), -10-5\rangle = \langle 5, 5, -15\rangle$; $\langle x, y, z\rangle = \langle 10, 2, -10\rangle + t\langle 5, 5, -15\rangle$

5. $\mathbf{a} = \langle 1-(-4), 1-1, -1-(-1)\rangle = \langle 5, 0, 0\rangle$; $\langle x, y, z\rangle = \langle 1, 1, -1\rangle + t\langle 5, 0, 0\rangle$

6. $\mathbf{a} = \langle 3-5/2, 2-1, 1-(-2)\rangle = \langle 1/2, 1, 3\rangle$; $\langle x, y, z\rangle = \langle 3, 2, 1\rangle + t\langle 1/2, 1, 3\rangle$

7. $\mathbf{a} = \langle 2-6, 3-(-1), 5-8\rangle = \langle -4, 4, -3\rangle$; $x = 2-4t$, $y = 3+4t$, $z = 5-3t$

8. $\mathbf{a} = \langle 2-0, 0-4, 0-9\rangle = \langle 2, -4, -9\rangle$; $x = 2+2t$, $y = -4t$, $z = -9t$

9. $\mathbf{a} = \langle 1-3, 0-(-2), 0-(-7)\rangle = \langle -2, 2, 7\rangle$; $x = 1-2t$, $y = 2t$, $z = 7t$

10. $\mathbf{a} = \langle 0-(-2), 0-4, 5-0\rangle = \langle 2, -4, 5\rangle$; $x = 2t$, $y = -4t$, $z = 5+5t$

11. $\mathbf{a} = \langle 4-(-6), 1/2-(-1/4), 1/3-1/6\rangle = \langle 10, 3/4, 1/6\rangle$; $x = 4+10t$, $y = \dfrac{1}{2}+\dfrac{3}{4}t$, $z = \dfrac{1}{3}+\dfrac{1}{6}t$

12. $\mathbf{a} = \langle -3-4, 7-(-8), 9-(-1)\rangle = \langle -7, 15, 10\rangle$; $x = -3-7t$, $y = 7+15t$, $z = 9+10t$

13. $a_1 = 10-1 = 9$, $a_2 = 14-4 = 10$, $a_3 = -2-(-9) = 7$; $\dfrac{x-10}{9} = \dfrac{y-14}{10} = \dfrac{z+2}{7}$

14. $a_1 = 1-2/3 = 1/3$, $a_2 = 3-0 = 3$, $a_3 = 1/4-(-1/4) = 1/2$; $\dfrac{x-1}{1/3} = \dfrac{y-3}{3} = \dfrac{z-1/4}{1/2}$

15. $a_1 = -7-4 = -11$, $a_2 = 2-2 = 0$, $a_3 = 5-1 = 4$; $\dfrac{x+7}{-11} = \dfrac{z-5}{4}$, $y = 2$

16. $a_1 = 1-(-5) = 6$, $a_2 = 1-(-2) = 3$, $a_3 = 2-(-4) = 6$; $\dfrac{x-1}{6} = \dfrac{y-1}{3} = \dfrac{z-2}{6}$

17. $a_1 = 5-5 = 0$, $a_2 = 10-1 = 9$, $a_3 = -2-(-14) = 12$; $x = 5$, $\dfrac{y-10}{9} = \dfrac{z+2}{12}$

18. $a_1 = 5/6-1/3 = 1/2$; $a_2 = -1/4-3/8 = -5/8$; $a_3 = 1/5-1/10 = 1/10$

$$\frac{x-5/6}{1/2} = \frac{y+1/4}{-5/8} = \frac{z-1/5}{1/10}$$

19. parametric: $x = 4+3t$, $y = 6+t/2$, $z = -7-3t/2$; symmetric: $\dfrac{x-4}{3} = \dfrac{y-6}{1/2} = \dfrac{z+7}{-3/2}$

20. parametric: $x = 1-7t$, $y = 8-8t$, $z = -2$; symmetric: $\dfrac{x-1}{-7} = \dfrac{y-8}{-8}$, $z = -2$

21. parametric: $x = 5t$, $y = 9t$, $z = 4t$; symmetric: $\dfrac{x}{5} = \dfrac{y}{9} = \dfrac{z}{4}$

22. parametric: $x = 12t$, $y = -3-5t$, $z = 10-6t$; symmetric: $\dfrac{x}{12} = \dfrac{y+3}{-5} = \dfrac{z-10}{-6}$

23. Writing the given line in the form $x/2 = (y-1)/(-3) = (z-5)/6$, we see that a direction vector is $\langle 2, -3, 6\rangle$. Parametric equations for the line are $x = 6+2t$, $y = 4-3t$, $z = -2+6t$.

24. A direction vector is $\langle 5, 1/3, -2\rangle$. Symmetric equations for the line are $(x-4)/5 = (y+11)/(1/3) = (z+7)/(-2)$.

25. A direction vector parallel to both the xz- and xy-planes is $\mathbf{i} = \langle 1, 0, 0\rangle$. Parametric equations for the line are $x = 2+t$, $y = -2$, $z = 15$.

26. **(a)** Since the unit vector $\mathbf{j} = \langle 0, 1, 0\rangle$ lies along the y-axis, we have $x = 1$, $y = 2+t$, $z = 8$.

(b) since the unit vector $\mathbf{k} = \langle 0, 0, 1\rangle$ is perpendicular to the xy-plane, we have $x = 1$, $y = 2$, $z = 8+t$.

27. Both lines go through the points $(0, 0, 0)$ and $(6, 6, 6)$. Since two points determine a line, the lines are the same.

28. $\mathbf{a}$ and $\mathbf{f}$ are parallel since $\langle 9, -12, 6\rangle = -3\langle -3, 4, -2\rangle$. $\mathbf{c}$ and $\mathbf{d}$ are orthogonal since $\langle 2, -3, 4\rangle \cdot \langle 1, 4, 5/2\rangle = 0$.

29. In the xy-plane, $z = 9+3t = 0$ and $t = -3$. Then $x = 4-2(-3) = 10$ and $y = 1+2(-3) = -5$. The point is $(10, -5, 0)$. In the xz-plane, $y = 1+2t = 0$ and $t = -1/2$. Then $x = 4-2(-1/2) = 5$ and $z = 9+3(-1/2) = 15/2$. The point is $(5, 0, 15/2)$. In the yz-plane, $x = 4-2t = 0$ and $t = 2$. Then $y = 1+2(2) = 5$ and $z = 9+3(2) = 15$. The point is $(0, 5, 15)$.

30. The parametric equations for the line are $x = 1+2t$, $y = -2+3t$, $z = 4+2t$. In the xy-plane, $z = 4+2t = 0$ and $t = -2$. Then $x = 1+2(-2) = -3$ and $y = -2+3(-2) = -8$. The point is $(-3, -8, 0)$. In the xz-plane, $y = -2+3t = 0$ and $t = 2/3$. Then $x = 1+2(2/3) = 7/3$ and $z = 4+2(2/3) = 16/3$. The point is $(7/3, 0, 16/3)$. In the yz-plane, $x = 1+2t = 0$ and $t = -1/2$. Then $y = -2+3(-1/2) = -7/2$ and $z = 4+2(-1/2) = 3$. The point is $(0, -7/2, 3)$.

31. Solving the system $4+t=6+2s$, $5+t=11+4s$, $-1+2t=-3+s$, or $t-2s=2$, $t-4s=6$, $2t-s=-2$ yields $s=-2$ and $t=-2$ in all three equations. Thus, the lines intersect at the point $x=4+(-2)=2$, $y=5+(-2)=3$, $z=-1+2(-2)=-5$, or $(2,3,-5)$.

32. Solving the system $1+t=2-s$, $2-t=1+s$, $3t=6s$, or $t+s=1$, $t+s=1$, $t-2s=0$ yields $s=1/3$ and $t=2/3$ in all three equations. Thus, the lines intersect at the point $x=1+2/3=5/3$, $y=2-2/3=4/3$, $z=3(2/3)=2$, or $(5/3,4/3,2)$.

33. The system of equations $2-t=4+s$, $3+t=1+s$, $1+t=1-s$, or $t+s=-2$, $t-s=-2$, $t+s=0$ has no solution since $-2\neq 0$. Thus, the lines do not intersect.

34. Solving the system $3-t=2+2s$, $2+t=-2+3s$, $8+2t=-2+8s$, or $t+2s=1$, $t-3s=-4$, $2t-8s=-10$ yields $s=1$ and $t=-1$ in all three equations. Thus, the lines intersect at the point $x=3-(-1)=4$, $y=2+(-1)=1$, $z=8+2(-1)=6$, or $(4,1,6)$.

35. $\mathbf{a}=\langle -1,2,-2\rangle$, $\mathbf{b}=\langle 2,3,-6\rangle$, $\mathbf{a}\cdot\mathbf{b}=16$, $\|\mathbf{a}\|=3$, $\|\mathbf{b}\|=7$; $\cos\theta=\dfrac{\mathbf{a}\cdot\mathbf{b}}{\|\mathbf{a}\|\,\|\mathbf{b}\|}=\dfrac{16}{3\cdot 7}$;

$\theta=\cos^{-1}\dfrac{16}{21}\approx 40.37^\circ$

36. $\mathbf{a}=\langle 2,7,-1\rangle$, $\mathbf{b}=\langle -2,1,4\rangle$, $\mathbf{a}\cdot\mathbf{b}=-1$, $\|\mathbf{a}\|=3\sqrt{6}$, $\|\mathbf{b}\|=\sqrt{21}$;

$\cos\theta=\dfrac{\mathbf{a}\cdot\mathbf{b}}{\|\mathbf{a}\|\,\|\mathbf{b}\|}=\dfrac{-1}{(3\sqrt{6})(\sqrt{21})}=-\dfrac{1}{9\sqrt{14}}$; $\theta=\cos^{-1}(-\dfrac{1}{9\sqrt{14}})\approx 91.70^\circ$

37. A direction vector perpendicular to the given lines will be $\langle 1,1,1\rangle\times\langle -2,1,-5\rangle=\langle -6,3,3\rangle$. Equations of the line are $x=4-6t$, $y=1+3t$, $z=6+3t$.

38. The direction vectors of the given lines are $\langle 3,2,4\rangle$ and $\langle 6,4,8\rangle=2\langle 3,2,4\rangle$. These are parallel, so we need a third vector parallel to the plane containing the lines which is not parallel to them. The point $(1,-1,0)$ is on the first line and $(-4,6,10)$ is on the second line. A third vector is then $\langle 1,-1,0\rangle-\langle -4,6,10\rangle=\langle 5,-7,-10\rangle$. Now a direction vector perpendicular to the plane is $\langle 3,2,4\rangle\times\langle 5,-7,-10\rangle=\langle 8,50,-31\rangle$. Equations of the line through $(1,-1,0)$ and perpendicular to the plane are $x=1+8t$, $y=-1+50t$, $z=-31t$.

39. $2(x-5)-3(y-1)+4(z-3)=0$; $2x-3y+4z=19$

40. $4(x-1)-2(y-2)+0(z-5)=0$; $4x-2y=0$

41. $-5(x-6)+0(y-10)+3(z+7)=0$; $-5x+3z=-51$

42. $6x-y+3z=0$

43. $6(x-1/2)+8(y-3/4)-4(z+1/2)=0$; $6x+8y-4z=11$

44. $-(x+1)+(y-1)-(z-0)=0$; $-x+y-z=2$

45. From the points $(3,5,2)$ and $(2,3,1)$ we obtain the vector $\mathbf{u}=\mathbf{i}+2\mathbf{j}+\mathbf{k}$. From the points $(2,3,1)$ and $(-1,-1,4)$ we obtain the vector $\mathbf{v}=3\mathbf{i}+4\mathbf{j}-3\mathbf{k}$. From the points $(-1,-1,4)$ and (x,y,z) we obtain the vector $\mathbf{w}=(x+1)\mathbf{i}+(y+1)\mathbf{j}+(z-4)\mathbf{k}$. Then, a normal vector is

$$\mathbf{u}\times\mathbf{v}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\ 1&2&1\\ 3&4&-3\end{vmatrix}=-10\mathbf{i}+6\mathbf{j}-2\mathbf{k}.$$

A vector equation of the plane is $-10(x+1)+6(y+1)-2(z-4)=0$ or $5x-3y+z=2$.

46. From the points $(0,1,0)$ and $(0,1,1)$ we obtain the vector $\mathbf{u}=\mathbf{k}$. From the points $(0,1,1)$ and $(1,3,-1)$ we obtain the vector $\mathbf{v}=\mathbf{i}+2\mathbf{j}-2\mathbf{k}$. From the points $(1,3,-1)$ and (x,y,z) we obtain the vector

$\mathbf{w} = (x-1)\mathbf{i} + (y-3)\mathbf{j} + (z+1)\mathbf{k}$. Then, a normal vector is

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 0 & 1 \\ 1 & 2 & -2 \end{vmatrix} = -2\mathbf{i} + \mathbf{j}.$$

A vector equation of the plane is $-2(x-1) + (y-3) + 0(z+1) = 0$ or $-2x + y = 1$.

47. From the points $(0,0,0)$ and $(1,1,1)$ we obtain the vector $\mathbf{u} = \mathbf{i} + \mathbf{j} + \mathbf{k}$. From the points $(1,1,1)$ and $(3,2,-1)$ we obtain the vector $\mathbf{v} = 2\mathbf{i} + \mathbf{j} - 2\mathbf{k}$. From the points $(3,2,-1)$ and (x,y,z) we obtain the vector $\mathbf{w} = (x-3)\mathbf{i} + (y-2)\mathbf{j} + (z+1)\mathbf{k}$. Then, a normal vector is

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 1 & 1 \\ 2 & 1 & -2 \end{vmatrix} = -3\mathbf{i} + 4\mathbf{j} - \mathbf{k}.$$

A vector equation of the plane is $-3(x-3) + 4(y-2) - (z+1) = 0$ or $-3x + 4y - z = 0$.

48. The three points are not colinear and all satisfy $x = 0$, which is the equation of the plane.

49. From the points $(1,2,-1)$ and $(4,3,1)$ we obtain the vector $\mathbf{u} = 3\mathbf{i} + \mathbf{j} + 2\mathbf{k}$. From the points $(4,3,1)$ and $(7,4,3)$ we obtain the vector $\mathbf{v} = 3\mathbf{i} + \mathbf{j} + 2\mathbf{k}$. From the points $(7,4,3)$ and (x,y,z) we obtain the vector $\mathbf{w} = (x-7)\mathbf{i} + (y-4)\mathbf{j} + (z-3)\mathbf{k}$. Since $\mathbf{u} \times \mathbf{v} = \mathbf{0}$, the points are colinear.

50. From the points $(2,1,2)$ and $(4,1,0)$ we obtain the vector $\mathbf{u} = 2\mathbf{i} - 2\mathbf{k}$. From the points $(4,1,0)$ and $(5,0,-5)$ we obtain the vector $\mathbf{v} = \mathbf{i} - \mathbf{j} - 5\mathbf{k}$. From the points $(5,0,-5)$ and (x,y,z) we obtain the vector $\mathbf{w} = (x-5)\mathbf{i} + y\mathbf{j} + (z+5)\mathbf{k}$. Then, a normal vector is

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 0 & -2 \\ 1 & -1 & -5 \end{vmatrix} = -2\mathbf{i} + 8\mathbf{j} - 2\mathbf{k}.$$

A vector equation of the plane is $-2(x-5) + 8y - 2(z+5) = 0$ or $x - 4y + z = 0$.

51. A normal vector to $x + y - 4z = 1$ is $\langle 1, 1, -4 \rangle$. The equation of the parallel plane is $(x-2) + (y-3) - 4(z+5) = 0$ or $x + y - 4z = 25$.

52. A normal vector to $5x - y + z = 6$ is $\langle 5, -1, 1, \rangle$. The equation of the parallel plane is $5(x-0) - (y-0) + (z-0) = 0$ or $5x - y + z = 0$.

53. A normal vector to the xy-plane is $\langle 0, 0, 1 \rangle$. The equation of the parallel plane is $z - 12 = 0$ or $z = 12$.

54. A normal vector is $\langle 0, 1, 0 \rangle$. The equation of the plane is $y + 5 = 0$ or $y = -5$.

55. Direction vectors of the lines are $\langle 3, -1, 1 \rangle$. and $\langle 4, 2, 1 \rangle$. A normal vector to the plane is $\langle 3, -1, 1 \rangle \times \langle 4, 2, 1 \rangle = \langle -3, 1, 10 \rangle$. A point on the first line, and thus in the plane, is $\langle 1, 1, 2 \rangle$. The equation of the plane is $-3(x-1) + (y-1) + 10(z-2) = 0$ or $-3x + y + 10z = 18$.

56. Direction vectors of the lines are $\langle 2, -1, 6 \rangle$ and $\langle 1, 1, -3 \rangle$. A normal vector to the plane is $\langle 2, -1, 6 \rangle \times \langle 1, 1, -3 \rangle = \langle -3, 12, 3 \rangle$. A point on the first line, and thus in the plane, is $(1, -1, 5)$. The equation of the plane is $-3(x-1) + 12(y+1) + 3(z-5) = 0$ or $-x + 4y + z = 0$.

57. A direction vector for the two lines is $\langle 1, 2, 1 \rangle$. Points on the lines are $(1,1,3)$ and $(3,0,-2)$. Thus, another vector parallel to the plane is $\langle 1-3, 1-0, 3+2 \rangle = \langle -2, 1, 5 \rangle$. A normal vector to the plane is $\langle 1, 2, 1 \rangle \times \langle -2, 1, 5 \rangle = \langle 9, -7, 5 \rangle$. Using the point $(3,0,-2)$ in the plane, the equation of the plane is $9(x-3) - 7(y-0) + 5(z+2) = 0$ or $9x - 7y + 5z = 17$.

58. A direction vector for the line is $\langle 3, 2, -2\rangle$. Letting $t = 0$, we see that the origin is on the line and hence in the plane. Thus, another vector parallel to the plane is $\langle 4-0, 0-0, -6-0\rangle = \langle 4, 0, -6\rangle$. A normal vector to the plane is $\langle 3, 2, -2\rangle \times \langle 4, 0, -6\rangle = \langle -12, 10, -8\rangle$. The equation of the plane is $-12(x-0)+10(y-0)-8(z-0) = 0$ or $6x - 5y + 4z = 0$.

59. A direction vector for the line, and hence a normal vector to the plane, is $\langle -3, 1, -1/2\rangle$. The equation of the plane is $-3(x-2) + (y-4) - \frac{1}{2}(z-8) = 0$ or $-3x + y - \frac{1}{2}z = -6$.

60. A normal vector to the plane is $\langle 2-1, 6-0, -3+2\rangle = \langle 1, 6, -1\rangle$. The equation of the plane is $(x-1) + 6(y-1) - (z-1) = 0$ or $x + 6y - z = 6$.

61. Normal vectors to the planes are **(a)** $\langle 2, -1, 3\rangle$, **(b)** $\langle 1, 2, 2\rangle$, **(c)** $\langle 1, 1, -3/2\rangle$, **(d)** $\langle -5, 2, 4\rangle$, **(e)** $\langle -8, -8, 12\rangle$, **(f)** $\langle -2, 1, -3\rangle$. Parallel planes are **(c)** and **(e)**, and **(a)** and **(f)**. Perpendicular planes are **(a)** and **(d)**, **(b)** and **(c)**, **(b)** and **(e)**, and **(d)** and **(f)**.

62. A normal vector to the plane is $\langle -7, 2, 3\rangle$. This is a direction vector for the line and the equations of the line are $x = -4 - 7t$, $y = 1 + 2t$, $z = 7 + 3t$.

63. A direction vector of the line is $\langle -6, 9, 3\rangle$, and the normal vectors of the planes are **(a)** $\langle 4, 1, 2\rangle$, **(b)** $\langle 2, -3, 1\rangle$, **(c)** $\langle 10, -15, -5\rangle$, **(d)** $\langle -4, 6, 2\rangle$. Vectors **(c)** and **(d)** are multiples of the direction vector and hence the corresponding planes are perpendicular to the line.

64. A direction vector of the line is $\langle -2, 4, 1\rangle$, and normal vectors to the planes are **(a)** $\langle 1, -1, 3\rangle$, **(b)** $\langle 6, -3, 0\rangle$, **(c)** $\langle 1, -2, 5\rangle$, **(d)** $\langle -2, 1, -2\rangle$. Since the dot product of each normal vector with the direction vector is non-zero, none of the planes are parallel to the line.

65. Letting $z = t$ in both equations and solving $5x - 4y = 8 + 9t$, $x + 4y = 4 - 3t$, we obtain $x = 2 + t$, $y = \frac{1}{2} - t$, $z = t$.

66. Letting $y = t$ in both equations and solving $x - z = 2 - 2t$, $3x + 2z = 1 + t$, we obtain $x = 1 - \frac{3}{5}t$, $y = t$, $z = -1 + \frac{7}{5}t$ or, letting $t = 5s$, $x = 1 - 3s$, $y = 5s$, $z = -1 + 7s$.

67. Letting $z = t$ in both equations and solving $4x - 2y = 1 + t$, $x + y = 1 - 2t$, we obtain $x = \frac{1}{2} - \frac{1}{2}t$, $y = \frac{1}{2} - \frac{3}{2}t$, $z = t$.

68. Letting $z = t$ and using $y = 0$ in the first equation, we obtain $x = -\frac{1}{2}t$, $y = 0$, $z = t$.

69. Substituting the parametric equations into the equation of the plane, we obtain $2(1+2t)-3(2-t)+2(-3t) = -7$ or $t = -3$. Letting $t = -3$ in the equation of the line, we obtain the point of intersection $(-5, 5, 9)$.

70. Substituting the parametric equations into the equation of the plane, we obtain $(3-2t)+(1+6t)+4(2-\frac{1}{2}t) = 12$ or $2t = 0$. Letting $t = 0$ in the equation of the line, we obtain the point of intersection $(3, 1, 2)$.

71. Substituting the parametric equations into the equation of the plane, we obtain $1 + 2 - (1 + t) = 8$ or $t = -6$. Letting $t = -6$ in the equation of the line, we obtain the point of intersection $(1, 2, -5)$.

72. Substituting the parametric equations into the equation of the plane, we obtain $4 + t - 3(2 + t) + 2(1 + 5t) = 0$ or $t = 0$. Letting $t = 0$ in the equation of the line, we obtain the point of intersection $(4, 2, 1)$.

In Problems 73 and 74, the cross product of the normal vectors to the two planes will be a vector parallel to both planes, and hence a direction vector for a line parallel to the two planes.

73. Normal vectors are $\langle 1, 1, -4\rangle$ and $\langle 2, -1, 1\rangle$. A direction vector is

$$\langle 1, 1, -4\rangle \times \langle 2, -1, 1\rangle = \langle -3, -9, -3\rangle = -3\langle 1, 3, 1\rangle.$$

Equations of the line are $x = 5 + t$, $y = 6 + 3t$, $z = -12 + t$.

74. Normal vectors are $\langle 2, 0, 1\rangle$ and $\langle -1, 3, 1\rangle$. A direction vector is

$$\langle 2, 0, 1\rangle \times \langle -1, 3, 1\rangle = \langle -3, -3, 6\rangle = -3\langle 1, 1, -2\rangle.$$

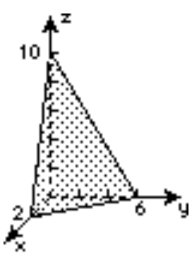
Equations of the line are $x = -3 + t$, $y = 5 + t$, $z = -1 - 2t$.

In Problems 75 and 76, the cross product of the direction vector of the line with the normal vector of the given plane will be a normal vector to the desired plane.

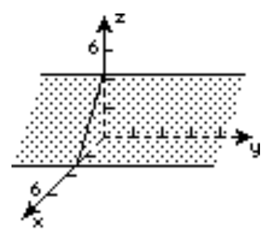
75. A direction vector of the line is $\langle 3, -1, 5\rangle$ and a normal vector to the given plane is $\langle 1, 1, 1\rangle$. A normal vector to the desired plane is $\langle 3, -1, 5\rangle \times \langle 1, 1, 1\rangle = \langle -6, 2, 4\rangle$. A point on the line, and hence in the plane, is $(4, 0, 1)$. The equation of the plane is $-6(x - 4) + 2(y - 0) + 4(z - 1) = 0$ or $3x - y - 2z = 10$.

76. A direction vector of the line is $\langle 3, 5, 2\rangle$ and a normal vector to the given plane is $\langle 2, -4, -1\rangle$. A normal vector to the desired plane is $\langle -3, 5, 2\rangle \times \langle 2, -4, -1\rangle = \langle 3, 1, 2\rangle$. A point on the line, and hence in the plane, is $(2, -2, 8)$. The equation of the plane is $3(x - 2) + (y + 2) + 2(z - 8) = 0$ or $3x + y + 2z = 20$.

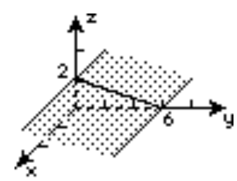
77.



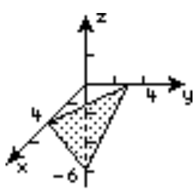
78.



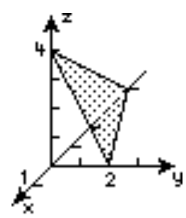
79.



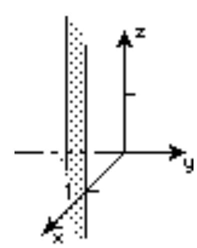
80.



81.



82.



EXERCISES 7.6

Vector Spaces

1. Not a vector space. Axiom (**vi**) is not satisfied.

2. Not a vector space. Axiom (**i**) is not satisfied.

3. Not a vector space. Axiom (**x**) is not satisfied.

4. A vector space

5. A vector space

6. A vector space

7. Not a vector space. Axiom (**ii**) is not satisfied.

8. A vector space

9. A vector space

10. Not a vector space. Axiom (**i**) is not satisfied.

11. A subspace

12. Not a subspace. Axiom (**i**) is not satisfied.

13. Not a subspace. Axiom (**ii**) is not satisfied.

14. A subspace

15. A subspace

16. A subspace

17. A subspace

18. A subspace

19. Not a subspace. Neither axioms (**i**) nor (**ii**) are satisfied.

20. A subspace

21. Let (x_1, y_1, z_1) and (x_2, y_2, z_2) be in S. Then

$$(x_1, y_1, z_1) + (x_2, y_2, z_2) = (at_1, bt_1, ct_1) + (at_2, bt_2, ct_2) = (a(t_1 + t_2), b(t_1 + t_2), c(t_1 + t_2))$$

is in S. Also, for (x, y, z) in S then $k(x, y, z) = (kx, ky, kz) = (a(kt), b(kt), c(kt))$ is also in S.

22. Let (x_1, y_1, z_1) and (x_2, y_2, z_2) be in S. Then $ax_1 + by_1 + cz_1 = 0$ and $ax_2 + by_2 + cz_2 = 0$. Adding gives $a(x_1 + x_2) + b(y_1 + y_2) + c(z_1 + z_2) = 0$ and so $(x_1, y_1, z_1) + (x_2, y_2, z_2) = (x_1 + x_2, y_1 + y_2, z_1 + z_2)$ is in S. Also, for (x, y, z) then $ax + by + cz = 0$ implies $k(ax + by + cz) = k \cdot 0 = 0$ and $a(kx) + b(ky) + c(kz) = 0$. this means $k(x, y, z) = (kx, ky, kz)$ is in S.

23. **(a)** $c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + c_3\mathbf{u}_3 = \mathbf{0}$ if and only if $c_1 + c_2 + c_3 = 0$, $c_2 + c_3 = 0$, $c_3 = 0$. The only solution of this system is $c_1 = 0$, $c_2 = 0$, $c_3 = 0$.

(b) Solving the system $c_1 + c_2 + c_3 = 3$, $c_2 + c_3 = -4$, $c_3 = 8$ gives $c_1 = 7$, $c_2 = -12$, $c_3 = 8$. Thus $\mathbf{a} = 7\mathbf{u}_1 - 12\mathbf{u}_2 + 8\mathbf{u}_3$.

24. **(a)** The assumption $c_1p_1 + c_2p_2 = 0$ is equivalent to $(c_1 + c_2)x + (c_1 - c_2) = 0$. Thus $c_1 + c_2 = 0$, $c_1 - c_2 = 0$. The only solution of this system is $c_1 = 0$, $c_2 = 0$.

(b) Solving the system $c_1 + c_2 = 5$, $c_1 - c_2 = 2$ gives $c_1 = \frac{7}{2}$, $c_2 = \frac{3}{2}$. Thus $p(x) = \frac{7}{2}p_1(x) + \frac{3}{2}p_2(x)$

25. Linearly dependent since $\langle -6, 12\rangle = -\frac{3}{2}\langle 4, -8\rangle$

26. Linearly dependent since $2\langle 1, 1\rangle + 3\langle 0, 1\rangle + (-1)\langle 2, 5\rangle = \langle 0, 0\rangle$

27. Linearly independent

28. Linearly dependent since for all x $(1)\cdot 1 + (-2)(x+1) + (1)(x+1)^2 + (-1)x^2 = 0$.

29. f is discontinuous at $x = -1$ and at $x = -3$.

30. $(x, \sin x) = \displaystyle\int_0^{2\pi} x\sin x\,dx = (-x\cos x + \sin x)\Big|_0^{2\pi} = -2\pi$

31. $\|x\|^2 = \displaystyle\int_0^{2\pi} x^2\,dx = \frac{1}{3}x^3\Big|_0^{2\pi} = \frac{8}{3}\pi^3$ and so $\|x\| = 2\sqrt{\dfrac{2\pi^3}{3}}$. Now

$$\|\sin x\|^2 = \int_0^{2\pi} \sin^2 x\,dx = \frac{1}{2}\int_0^{2\pi}(1-\cos 2x)\,dx = \frac{1}{2}\left(x - \frac{1}{2}\sin 2x\right)\Big|_0^{2\pi} = \pi$$

and so $\|\sin x\| = \sqrt{\pi}$.

32. A basis could be 1, x, $e^x\cos 3x$, $e^x\sin 3x$.

33. We need to show that $\text{Span}\{x_1, x_2, \ldots, x_n\}$ is closed under vector addition and scalar multiplication. Suppose $\mathbf{u}$ and $\mathbf{v}$ are in $\text{Span}\{x_1, x_2, \ldots, x_n\}$. Then $\mathbf{u} = a_1x_1 + a_2x_2 + \cdots + a_nx_n$ and $\mathbf{v} = b_1x_1 + b_2x_2 + \cdots + b_nx_n$, so that

$$\mathbf{u} + \mathbf{v} = (a_1+b_1)x_1 + (a_2+b_2)x_2 + \cdots + (a_n+b_n)x_n,$$

which is in $\text{Span}\{x_1, x_2, \ldots, x_n\}$. Also, for any real number k,

$$k\mathbf{u} = k(a_1x_1 + a_2x_2 + \cdots + a_nx_n) = ka_1x_1 + ka_2x_2 + \cdots + ka_nx_n,$$

which is in $\text{Span}\{x_1, x_2, \ldots, x_n\}$. Thus, $\text{Span}\{x_1, x_2, \ldots, x_n\}$ is a subspace of $\mathbf{V}$.

34. R^2 is not a subspace of either R^3 or R^4 and R^3 is not a subspace of R^4. The vectors in R^2 are ordered pairs, while the vectors in R^3 are ordered triples.

35. Since a basis for M_{22} is

$$B = \left\{\begin{bmatrix}1 & 0\\ 0 & 0\end{bmatrix}, \begin{bmatrix}0 & 1\\ 0 & 0\end{bmatrix}, \begin{bmatrix}0 & 0\\ 1 & 0\end{bmatrix}, \begin{bmatrix}0 & 0\\ 0 & 1\end{bmatrix}\right\},$$

the dimension of M_{22} is 4.

36. To show that the set of nonzero orthogonal vectors is linearly independent we set $c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_n\mathbf{v}_n = \mathbf{0}$. For $0 \le i \le n$,

$$(c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_i\mathbf{v}_i \cdots + c_n\mathbf{v}_n)\cdot\mathbf{v}_i = c_1\mathbf{v}_1\cdot\mathbf{v}_i + c_2\mathbf{v}_2\cdot\mathbf{v}_i + \cdots + c_i\mathbf{v}_i\cdot\mathbf{v}_i \cdots + c_n\mathbf{v}_n\cdot\mathbf{v}_i = c_i||\mathbf{v}_i||^2,$$

so $c_i||\mathbf{v}_i||^2 = 0$ because

$$(c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_i\mathbf{v}_i \cdots + c_n\mathbf{v}_n)\cdot\mathbf{v}_i = \mathbf{0}\cdot\mathbf{v}_i = 0.$$

Since $\mathbf{v}_i$ is a nonzero vector, $c_i = 0$. Thus, the assumption that $c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_n\mathbf{v}_n = \mathbf{0}$ leads to $c_1 = c_2 = \cdots = c_n = 0$, and the set is linearly independent.

37. We verify the four properties:

(i) $(\mathbf{u}, \mathbf{v}) = u_1v_1 + 4u_2v_2 = v_1u_1 + 4v_2u_2 = (\mathbf{v}, \mathbf{u})$

(ii) $(k\mathbf{u}, \mathbf{v}) = (ku_1)v_1 + 4(ku_2)v_2 = k(u_1v_1 + 4u_2v_2) = k(\mathbf{u}, \mathbf{v})$

(iii) $(\mathbf{u}, \mathbf{u}) = u_1^2 + 4ku_2^2 > 0$ for $\mathbf{u} \ne \mathbf{0}$. Furthermore, $u_1^2 + 4ku_2^2 = 0$ if and only if $u_1 = 0$ and $u_2 = 0$, or equivalently, $\mathbf{u} = \mathbf{0}$.

(iv) $(\mathbf{u}, \mathbf{v} + \mathbf{w}) = u_1(v_1 + w_1) + 4u_2(v_2 + w_2) = (u_1v_1 + 4u_2v_2) + (u_1w_1 + 4u_2w_2) = (\mathbf{u}, \mathbf{v}) + (\mathbf{u}, \mathbf{w})$

38. **(a)** Let $\mathbf{u} = \langle 2, 1\rangle$ and $\mathbf{v} = \langle 2, -1\rangle$ be nonzero vectors in R^2. With respect to the standard inner or dot product on R^2,

$$\mathbf{u}\cdot\mathbf{v} = \langle 2,1\rangle\cdot\langle 2,-1\rangle = 2\cdot 2 + 1\cdot(-1) = 3.$$

We see that $\mathbf{u}$ and $\mathbf{v}$ are not orthogonal with respect to that inner product. But using the inner product in Problem 37, we have

$$(\mathbf{u},\mathbf{v}) = 2\cdot 2 + 4(1)\cdot(-1) = 0,$$

and so $\mathbf{u}$ and $\mathbf{v}$ are orthogonal with respect to that inner product.

(b) Consider $f(x) = \sin x$ and $g(x) = \cos x$ in $C[0, 2\pi]$. Since

$$\int_0^{2\pi} \sin x\cos x\,dx = \frac{1}{2}\int_0^{2\pi}\sin 2x\,dx = -\frac{1}{4}\cos 2x\Big|_0^{2\pi} = -\frac{1}{4}(1-1) = 0,$$

these functions are orthogonal in $C[0, 2\pi]$.

EXERCISES 7.7

Gram-Schmidt Orthogonalization Process

1. Letting $\mathbf{w}_1 = \langle \frac{12}{13}, \frac{5}{13}\rangle$ and $\mathbf{w}_2 = \langle \frac{5}{13}, -\frac{12}{13}\rangle$, we have

$$\mathbf{w}_1\cdot\mathbf{w}_2 = \left(\frac{12}{13}\right)\left(\frac{5}{13}\right) + \left(\frac{5}{13}\right)\left(-\frac{12}{13}\right) = 0,$$

so the vectors are orthogonal. Also,

$$||\mathbf{w}_1|| = \sqrt{\left(\frac{12}{13}\right)^2 + \left(\frac{5}{13}\right)^2} = 1 \quad\text{and}\quad ||\mathbf{w}_2|| = \sqrt{\left(\frac{5}{13}\right)^2 + \left(-\frac{12}{13}\right)^2} = 1,$$

so the basis is orthonormal. To express $\mathbf{u} = \langle 4, 2\rangle$ in terms of $\mathbf{w}_1$ and $\mathbf{w}_2$ we compute

$$\mathbf{u}\cdot\mathbf{w}_1 = \langle 4,2\rangle\cdot\left\langle\frac{12}{13},\frac{5}{13}\right\rangle = (4)\left(\frac{12}{13}\right) + (2)\left(\frac{5}{13}\right) = \frac{58}{13}$$
$$\mathbf{u}\cdot\mathbf{w}_2 = \langle 4,2\rangle\cdot\left\langle\frac{5}{13},-\frac{12}{13}\right\rangle = (4)\left(\frac{5}{13}\right) + (2)\left(-\frac{12}{13}\right) = -\frac{4}{13},$$

so

$$\mathbf{u} = \frac{58}{13}\mathbf{w}_1 - \frac{4}{13}\mathbf{w}_2.$$

2. Letting $\mathbf{w}_1 = \langle 1/\sqrt{3}, 1/\sqrt{3}, -1/\sqrt{3}\rangle$, $\mathbf{w}_2 = \langle 0, -1/\sqrt{2}, -1/\sqrt{2}\rangle$, and $\mathbf{w}_3 = \langle -2/\sqrt{6}, 1/\sqrt{6}, -1/\sqrt{6}\rangle$, we have

$$\mathbf{w}_1\cdot\mathbf{w}_2 = \left(\frac{1}{\sqrt{3}}\right)(0) + \left(\frac{1}{\sqrt{3}}\right)\left(-\frac{1}{\sqrt{2}}\right) + \left(-\frac{1}{\sqrt{3}}\right)\left(-\frac{1}{\sqrt{2}}\right) = 0$$
$$\mathbf{w}_1\cdot\mathbf{w}_3 = \left(\frac{1}{\sqrt{3}}\right)\left(-\frac{2}{\sqrt{6}}\right) + \left(\frac{1}{\sqrt{3}}\right)\left(\frac{1}{\sqrt{6}}\right) + \left(-\frac{1}{\sqrt{3}}\right)\left(-\frac{1}{\sqrt{6}}\right) = 0$$
$$\mathbf{w}_2\cdot\mathbf{w}_3 = (0)\left(-\frac{2}{\sqrt{6}}\right) + \left(-\frac{1}{\sqrt{2}}\right)\left(\frac{1}{\sqrt{6}}\right) + \left(-\frac{1}{\sqrt{2}}\right)\left(-\frac{1}{\sqrt{6}}\right) = 0,$$

so the vectors are orthogonal. Also,

$$||\mathbf{w}_1|| = \sqrt{\left(\frac{1}{\sqrt{3}}\right)^2 + \left(\frac{1}{\sqrt{3}}\right)^2 + \left(-\frac{1}{\sqrt{3}}\right)^2} = 1, \quad ||\mathbf{w}_2|| = \sqrt{0^2 + \left(-\frac{1}{\sqrt{2}}\right)^2 + \left(-\frac{1}{\sqrt{2}}\right)^2} = 1,$$

$$\text{and} \quad ||\mathbf{w}_3|| = \sqrt{\left(-\frac{2}{\sqrt{6}}\right)^2 + \left(\frac{1}{\sqrt{6}}\right)^2 + \left(-\frac{1}{\sqrt{6}}\right)^2} = 1,$$

so the basis is orthonormal. To express $\mathbf{u} = \langle 5, -1, 6\rangle$ in terms of $\mathbf{w}_1$, $\mathbf{w}_2$, and $\mathbf{w}_3$ we compute

$$\mathbf{u}\cdot\mathbf{w}_1 = \langle 5,-1,6\rangle \cdot \left\langle \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}\right\rangle = (5)\left(\frac{1}{\sqrt{3}}\right) + (-1)\left(\frac{1}{\sqrt{3}}\right) + (6)\left(-\frac{1}{\sqrt{3}}\right) = -\frac{2}{\sqrt{3}}$$

$$\mathbf{u}\cdot\mathbf{w}_2 = \langle 5,-1,6\rangle \cdot \left\langle 0, -\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right\rangle = (5)(0) + (-1)\left(-\frac{1}{\sqrt{2}}\right) + (6)\left(-\frac{1}{\sqrt{2}}\right) = -\frac{5}{\sqrt{2}}$$

$$\mathbf{u}\cdot\mathbf{w}_3 = \langle 5,-1,6\rangle \cdot \left\langle -\frac{2}{\sqrt{6}}, \frac{1}{\sqrt{6}}, -\frac{1}{\sqrt{6}}\right\rangle = (5)\left(-\frac{2}{\sqrt{6}}\right) + (-1)\left(\frac{1}{\sqrt{6}}\right) + (6)\left(-\frac{1}{\sqrt{6}}\right) = -\frac{17}{\sqrt{6}}$$

so

$$\mathbf{u} = -\frac{2}{\sqrt{3}}\,\mathbf{w}_1 - \frac{5}{\sqrt{2}}\,\mathbf{w}_2 - \frac{17}{\sqrt{6}}\,\mathbf{w}_3.$$

Since the basis vectors in Problems 3 *and* 4 *are orthogonal but not orthonormal, the result of Theorem* 7.5 *must be slightly modified to read*

$$\mathbf{u} = \frac{\mathbf{u}\cdot\mathbf{w}_1}{||\mathbf{w}_1||^2}\,\mathbf{w}_1 + \frac{\mathbf{u}\cdot\mathbf{w}_2}{||\mathbf{w}_2||^2}\,\mathbf{w}_2 + \cdots + \frac{\mathbf{u}\cdot\mathbf{w}_n}{||\mathbf{w}_n||^2}\,\mathbf{w}_n.$$

The proof is very similar to that given in the text for Theorem 7.5*.*

3. Letting $\mathbf{w}_1 = \langle 1,0,1\rangle, \mathbf{w}_2 = \langle 0,1,0\rangle$, and $\mathbf{w}_3 = \langle -1,0,1\rangle$ we have

$$\mathbf{w}_1\cdot\mathbf{w}_2 = (1)(0) + (0)(1) + (1)(0) = 0$$
$$\mathbf{w}_1\cdot\mathbf{w}_3 = (1)(-1) + (0)(0) + (1)(1) = 0$$
$$\mathbf{w}_2\cdot\mathbf{w}_3 = (0)(-1) + (1)(0) + (0)(1) = 0$$

so the vectors are orthogonal. We also compute

$$||\mathbf{w}_1||^2 = 1^2 + 0^2 + 1^2 = 2$$
$$||\mathbf{w}_2||^2 = 0^2 + 1^2 + 0^2 = 1$$
$$||\mathbf{w}_3||^2 = (-1)^2 + 0^2 + 1^2 = 2$$

and, with $\mathbf{u} = \langle 10, 7, -13\rangle$,

$$\mathbf{u}\cdot\mathbf{w}_1 = (10)(1) + (7)(0) + (-13)(1) = -3$$
$$\mathbf{u}\cdot\mathbf{w}_2 = (10)(0) + (7)(1) + (-13)(0) = 7$$
$$\mathbf{u}\cdot\mathbf{w}_3 = (10)(-1) + (7)(0) + (-13)(1) = -23.$$

Then, using the result given before the solution to this problem, we have

$$\mathbf{u} = -\frac{3}{2}\,\mathbf{w}_1 + 7\mathbf{w}_2 - \frac{23}{2}\,\mathbf{w}_3.$$

4. Letting $\mathbf{w}_1 = \langle 2, 1, -2, 0\rangle, \mathbf{w}_2 = \langle 1, 2, 2, 1\rangle$, $\mathbf{w}_3 = \langle 3, -4, 1, 3\rangle$, and $\mathbf{w}_4 = \langle 5, -2, 4, -9\rangle$ we have

$$\begin{aligned}
\mathbf{w}_1 \cdot \mathbf{w}_2 &= (2)(1) + (1)(2) + (-2)(2) + (0)(1) = 0\\
\mathbf{w}_1 \cdot \mathbf{w}_3 &= (2)(3) + (1)(-4) + (-2)(1) + (0)(3) = 0\\
\mathbf{w}_1 \cdot \mathbf{w}_4 &= (2)(5) + (1)(-2) + (-2)(4) + (0)(-9) = 0\\
\mathbf{w}_2 \cdot \mathbf{w}_3 &= (1)(3) + (2)(-4) + (2)(1) + (1)(3) = 0\\
\mathbf{w}_2 \cdot \mathbf{w}_4 &= (1)(5) + (2)(-2) + (2)(4) + (1)(-9) = 0\\
\mathbf{w}_3 \cdot \mathbf{w}_4 &= (3)(5) + (-4)(-2) + (1)(4) + (3)(-9) = 0
\end{aligned}$$

so the vectors are orthogonal. We also compute

$$\begin{aligned}
||\mathbf{w}_1||^2 &= 2^2 + 1^2 + (-2)^2 + 0^2 = 9\\
||\mathbf{w}_2||^2 &= 1^2 + 2^2 + 2^2 + 1^2 = 10\\
||\mathbf{w}_3||^2 &= 3^2 + (-4)^2 + 1^2 + 3^2 = 35\\
||\mathbf{w}_4||^2 &= 5^2 + (-2)^2 + 4^2 + (-9)^2 = 126
\end{aligned}$$

and, with $\mathbf{u} = \langle 1, 2, 4, 3\rangle$,

$$\begin{aligned}
\mathbf{u} \cdot \mathbf{w}_1 &= (1)(2) + (2)(1) + (4)(-2) + (3)(0) = -4\\
\mathbf{u} \cdot \mathbf{w}_2 &= (1)(1) + (2)(2) + (4)(2) + (3)(1) = 16\\
\mathbf{u} \cdot \mathbf{w}_3 &= (1)(3) + (2)(-4) + (4)(1) + (3)(3) = 8\\
\mathbf{u} \cdot \mathbf{w}_4 &= (1)(5) + (2)(-2) + (4)(4) + (3)(-9) = -10.
\end{aligned}$$

Then, using the result given before the solution to this problem, we have

$$\mathbf{u} = -\frac{4}{9}\,\mathbf{w}_1 + \frac{8}{5}\mathbf{w}_2 + \frac{8}{35}\,\mathbf{w}_3 - \frac{5}{63}\,\mathbf{w}_4.$$

5. (a) We have $\mathbf{u}_1 = \langle -3, 2\rangle$ and $\mathbf{u}_2 = \langle -1, -1\rangle$. Taking $\mathbf{v}_1 = \mathbf{u}_1 = \langle -3, 2\rangle$, and using $\mathbf{u}_2 \cdot \mathbf{v}_1 = 1$ and $\mathbf{v}_1 \cdot \mathbf{v}_1 = 13$ we obtain

$$\mathbf{v}_2 = \mathbf{u}_2 - \frac{\mathbf{u}_2 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1}\,\mathbf{v}_1 = \langle -1, -1\rangle - \frac{1}{13}\langle -3, 2\rangle = \left\langle -\frac{10}{13}, -\frac{15}{13}\right\rangle.$$

Thus, an orthogonal basis is $\{\langle -3, 2\rangle, \langle -\frac{10}{13}, -\frac{15}{13}\rangle\}$ and an orthonormal basis is $\{\mathbf{w}_1', \mathbf{w}_2'\}$, where

$$\mathbf{w}_1' = \frac{1}{||\langle -3, 2\rangle||}\,\langle -3, 2\rangle = \frac{1}{\sqrt{13}}\,\langle -3, 2\rangle = \left\langle -\frac{3}{\sqrt{13}}, \frac{2}{\sqrt{13}}\right\rangle$$

and

$$\mathbf{w}_2' = \frac{1}{||\langle -\frac{10}{13}, -\frac{15}{13}\rangle||}\,\left\langle -\frac{10}{13}, -\frac{15}{13}\right\rangle = \frac{1}{5/\sqrt{13}}\,\left\langle -\frac{10}{13}, -\frac{15}{13}\right\rangle = \left\langle -\frac{2}{\sqrt{13}}, -\frac{3}{\sqrt{13}}\right\rangle.$$

(b) We have $\mathbf{u}_1 = \langle -3, 2\rangle$ and $\mathbf{u}_2 = \langle -1, -1\rangle$. Taking $\mathbf{v}_1 = \mathbf{u}_2 = \langle -1, -1\rangle$, and using $\mathbf{u}_1 \cdot \mathbf{v}_1 = 1$ and $\mathbf{v}_1 \cdot \mathbf{v}_1 = 2$ we obtain

$$\mathbf{v}_2 = \mathbf{u}_1 - \frac{\mathbf{u}_1 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1}\,\mathbf{v}_1 = \langle -3, 2\rangle - \frac{1}{2}\langle -1, -1\rangle = \left\langle -\frac{5}{2}, \frac{5}{2}\right\rangle.$$

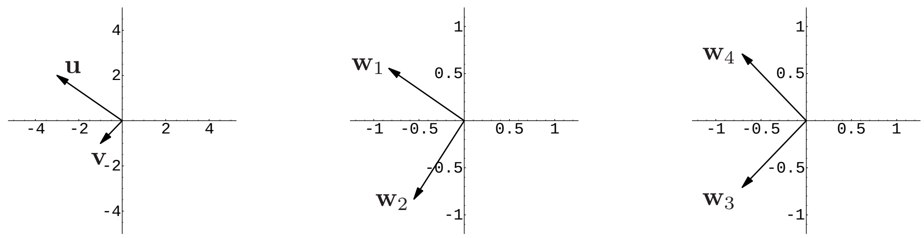
Thus, an orthogonal basis is $\{\langle -1, -1\rangle, \langle -\frac{5}{2}, \frac{5}{2}\rangle\}$ and an orthonormal basis is $\{\mathbf{w}_3'', \mathbf{w}_4''\}$, where

$$\mathbf{w}_3'' = \frac{1}{||\langle -1, -1\rangle||}\,\langle -1, -1\rangle = \frac{1}{\sqrt{2}}\,\langle -1, -1\rangle = \left\langle -\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right\rangle$$

and

$$\mathbf{w}_4'' = \frac{1}{||\langle -\frac{5}{2}, \frac{5}{2}\rangle||}\,\left\langle -\frac{5}{2}, \frac{5}{2}\right\rangle = \frac{1}{5/\sqrt{2}}\,\left\langle -\frac{5}{2}, \frac{5}{2}\right\rangle = \left\langle -\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right\rangle.$$

(c)



6. **(a)** We have $\mathbf{u}_1 = \langle -3, 4\rangle$ and $\mathbf{u}_2 = \langle -1, 0\rangle$. Taking $\mathbf{v}_1 = \mathbf{u}_1 = \langle -3, 4\rangle$, and using $\mathbf{u}_2 \cdot \mathbf{v}_1 = 3$ and $\mathbf{v}_1 \cdot \mathbf{v}_1 = 25$ we obtain

$$\mathbf{v}_2 = \mathbf{u}_2 - \frac{\mathbf{u}_2 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1}\,\mathbf{v}_1 = \langle -1, 0\rangle - \frac{3}{25}\langle -3, 4\rangle = \left\langle -\frac{16}{25}, -\frac{12}{25}\right\rangle.$$

Thus, an orthogonal basis is $\{\langle -3, 4\rangle, \langle -\frac{16}{25}, -\frac{12}{25}\rangle\}$ and an orthonormal basis is $\{\mathbf{w}_1', \mathbf{w}_2'\}$, where

$$\mathbf{w}_1' = \frac{1}{||\langle -3, 4\rangle||}\,\langle -3, 4\rangle = \frac{1}{5}\,\langle -3, 4\rangle = \left\langle -\frac{3}{5}, \frac{4}{5}\right\rangle$$

and

$$\mathbf{w}_2' = \frac{1}{||\langle -\frac{16}{25}, -\frac{12}{25}\rangle||}\,\left\langle -\frac{16}{25}, -\frac{12}{25}\right\rangle = \frac{1}{4/5}\,\left\langle -\frac{16}{25}, -\frac{12}{25}\right\rangle = \left\langle -\frac{4}{5}, -\frac{3}{5}\right\rangle.$$

(b) We have $\mathbf{u}_1 = \langle -3, 4\rangle$ and $\mathbf{u}_2 = \langle -1, 0\rangle$. Taking $\mathbf{v}_1 = \mathbf{u}_2 = \langle -1, 0\rangle$, and using $\mathbf{u}_1 \cdot \mathbf{v}_1 = 3$ and $\mathbf{v}_1 \cdot \mathbf{v}_1 = 1$ we obtain

$$\mathbf{v}_2 = \mathbf{u}_1 - \frac{\mathbf{u}_1 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1}\,\mathbf{v}_1 = \langle -3, 4\rangle - \frac{3}{1}\langle -1, 0\rangle = \langle 0, 4\rangle.$$

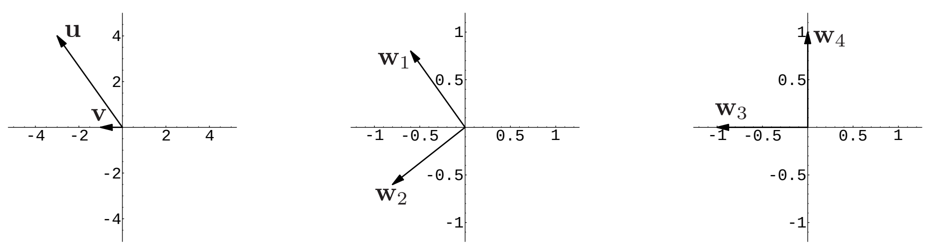
Thus, an orthogonal basis is $\{\langle -1, 0\rangle, \langle 0, 4\rangle\}$ and an orthonormal basis is $\{\mathbf{w}_3'', \mathbf{w}_4''\}$, where

$$\mathbf{w}_3'' = \frac{1}{||\langle -1, 0\rangle||}\,\langle -1, 0\rangle = \frac{1}{1}\,\langle -1, 0\rangle = \langle -1, 0\rangle$$

and

$$\mathbf{w}_4'' = \frac{1}{||\langle 0, 4\rangle||}\,\langle 0, 4\rangle = \frac{1}{4}\,\langle 0, 4\rangle = \langle 0, 1\rangle.$$

(c)



7. **(a)** We have $\mathbf{u}_1 = \langle 1, 1\rangle$ and $\mathbf{u}_2 = \langle 1, 0\rangle$. Taking $\mathbf{v}_1 = \mathbf{u}_1 = \langle 1, 1\rangle$, and using $\mathbf{u}_2 \cdot \mathbf{v}_1 = 1$ and $\mathbf{v}_1 \cdot \mathbf{v}_1 = 2$ we obtain

$$\mathbf{v}_2 = \mathbf{u}_2 - \frac{\mathbf{u}_2 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1}\,\mathbf{v}_1 = \langle 1, 0\rangle - \frac{1}{2}\langle 1, 1\rangle = \left\langle \frac{1}{2}, -\frac{1}{2}\right\rangle.$$

Thus, an orthogonal basis is $\{\langle 1, 1\rangle, \langle \frac{1}{2}, -\frac{1}{2}\rangle\}$ and an orthonormal basis is $\{\mathbf{w}_1', \mathbf{w}_2'\}$, where

$$\mathbf{w}_1' = \frac{1}{||\langle 1, 1\rangle||}\,\langle 1, 1\rangle = \frac{1}{\sqrt{2}}\,\langle 1, 1\rangle = \left\langle \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right\rangle$$

and

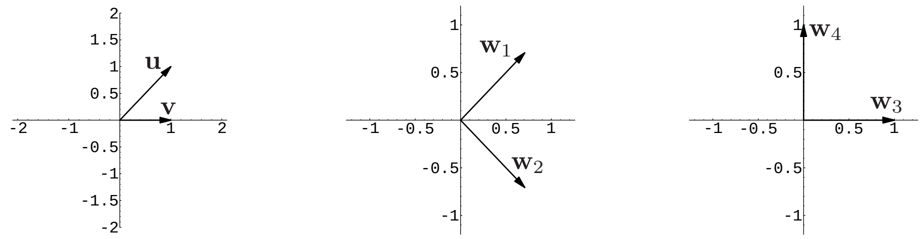
$$\mathbf{w}_2' = \frac{1}{||\langle \frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\rangle||}\,\left\langle \frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right\rangle = \frac{1}{1}\,\left\langle \frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right\rangle = \left\langle \frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right\rangle.$$

(b) We have $\mathbf{u}_1 = \langle 1, 1\rangle$ and $\mathbf{u}_2 = \langle 1, 0\rangle$. Taking $\mathbf{v}_1 = \mathbf{u}_2 = \langle 1, 0\rangle$, and using $\mathbf{u}_1 \cdot \mathbf{v}_1 = 1$ and $\mathbf{v}_1 \cdot \mathbf{v}_1 = 1$ we obtain

$$\mathbf{v}_2 = \mathbf{u}_1 - \frac{\mathbf{u}_1 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1}\,\mathbf{v}_1 = \langle 1, 1\rangle - \frac{1}{1}\langle 1, 0\rangle = \langle 0, 1\rangle\,.$$

Thus, an orthogonal basis is $\{\langle 1, 0\rangle, \langle 0, 1\rangle\}$, which is also an orthonormal basis.

(c)



8. **(a)** We have $\mathbf{u}_1 = \langle 5, 7\rangle$ and $\mathbf{u}_2 = \langle 1, -2\rangle$. Taking $\mathbf{v}_1 = \mathbf{u}_1 = \langle 5, 7\rangle$, and using $\mathbf{u}_2 \cdot \mathbf{v}_1 = -9$ and $\mathbf{v}_1 \cdot \mathbf{v}_1 = 74$ we obtain

$$\mathbf{v}_2 = \mathbf{u}_2 - \frac{\mathbf{u}_2 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1}\,\mathbf{v}_1 = \langle 1, -2\rangle - \frac{9}{74}\langle 5, 7\rangle = \left\langle \frac{119}{74}, -\frac{85}{74}\right\rangle.$$

Thus, an orthogonal basis is $\{\langle 5, 7\rangle, \langle \frac{119}{74}, -\frac{85}{74}\rangle\}$ and an orthonormal basis is $\{\mathbf{w}'_1, \mathbf{w}'_2\}$, where

$$\mathbf{w}'_1 = \frac{1}{||\langle 5, 7\rangle||}\,\langle 5, 7\rangle = \frac{1}{\sqrt{74}}\,\langle 5, 7\rangle = \left\langle \frac{5}{\sqrt{74}}, \frac{7}{\sqrt{74}}\right\rangle$$

and

$$\mathbf{w}'_2 = \frac{1}{||\langle \frac{119}{74}, -\frac{85}{74}\rangle||}\,\left\langle \frac{119}{74}, -\frac{85}{74}\right\rangle = \frac{1}{17/\sqrt{74}}\,\left\langle \frac{119}{74}, -\frac{85}{74}\right\rangle = \left\langle \frac{7}{\sqrt{74}}, -\frac{5}{\sqrt{74}}\right\rangle.$$

(b) We have $\mathbf{u}_1 = \langle 5, 7\rangle$ and $\mathbf{u}_2 = \langle 1, -2\rangle$. Taking $\mathbf{v}_1 = \mathbf{u}_2 = \langle 1, -2\rangle$, and using $\mathbf{u}_1 \cdot \mathbf{v}_1 = -9$ and $\mathbf{v}_1 \cdot \mathbf{v}_1 = 5$ we obtain

$$\mathbf{v}_2 = \mathbf{u}_1 - \frac{\mathbf{u}_1 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1}\,\mathbf{v}_1 = \langle 5, 7\rangle - \frac{9}{5}\langle 1, -2\rangle = \left\langle \frac{34}{5}, \frac{17}{5}\right\rangle.$$

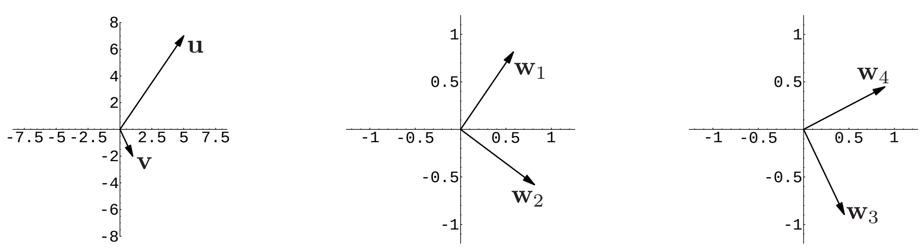
Thus, an orthogonal basis is $\{\langle 1, -2\rangle, \langle \frac{34}{5}, \frac{17}{5}\rangle\}$ and an orthonormal basis is $\{\mathbf{w}''_3, \mathbf{w}''_4\}$, where

$$\mathbf{w}''_3 = \frac{1}{||\langle 1, -2\rangle||}\,\langle 1, -2\rangle = \frac{1}{\sqrt{5}}\,\langle 1, -2\rangle = \left\langle \frac{1}{\sqrt{5}}, -\frac{2}{\sqrt{5}}\right\rangle$$

and

$$\mathbf{w}''_4 = \frac{1}{||\langle \frac{34}{5}, \frac{17}{5}\rangle||}\,\left\langle \frac{34}{5}, \frac{17}{5}\right\rangle = \frac{1}{17/\sqrt{5}}\,\left\langle \frac{34}{5}, \frac{17}{5}\right\rangle = \left\langle \frac{2}{\sqrt{5}}, \frac{1}{\sqrt{5}}\right\rangle.$$

(c)



9. We have $\mathbf{u}_1 = \langle 1, 1, 0\rangle, \mathbf{u}_2 = \langle 1, 2, 2\rangle$, and $\mathbf{u}_3 = \langle 2, 2, 1\rangle$. Taking $\mathbf{v}_1 = \mathbf{u}_1 = \langle 1, 1, 0\rangle$ and using $\mathbf{u}_2 \cdot \mathbf{v}_1 = 3$ and $\mathbf{v}_1 \cdot \mathbf{v}_1 = 2$ we obtain

$$\mathbf{v}_2 = \mathbf{u}_2 - \frac{\mathbf{u}_2 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1}\,\mathbf{v}_1 = \langle 1, 2, 2\rangle - \frac{3}{2}\langle 1, 1, 0\rangle = \left\langle -\frac{1}{2}, \frac{1}{2}, 2\right\rangle.$$

Next, using $\mathbf{u}_3 \cdot \mathbf{v}_1 = 4, \mathbf{u}_3 \cdot \mathbf{v}_2 = 2$, and $\mathbf{v}_2 \cdot \mathbf{v}_2 = \frac{9}{2}$, we obtain

$$\mathbf{v}_3 = \mathbf{u}_3 - \frac{\mathbf{u}_3 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1}\mathbf{v}_1 - \frac{\mathbf{u}_3 \cdot \mathbf{v}_2}{\mathbf{v}_2 \cdot \mathbf{v}_2}\mathbf{v}_2 = \langle 2,2,1\rangle - \frac{4}{2}\langle 1,1,0\rangle - \frac{2}{9/2}\left\langle -\frac{1}{2},\frac{1}{2},2\right\rangle = \left\langle \frac{2}{9},-\frac{2}{9},\frac{1}{9}\right\rangle.$$

Thus, an orthogonal basis is

$$B' = \left\{\langle 1,1,0\rangle, \left\langle -\frac{1}{2},\frac{1}{2},2\right\rangle, \left\langle \frac{2}{9},-\frac{2}{9},\frac{1}{9}\right\rangle\right\},$$

and an orthonormal basis is

$$B'' = \left\{\left\langle \frac{1}{\sqrt{2}},\frac{1}{\sqrt{2}},0\right\rangle, \left\langle -\frac{1}{3\sqrt{2}},\frac{1}{3\sqrt{2}},\frac{4}{3\sqrt{2}}\right\rangle, \left\langle \frac{2}{3},-\frac{2}{3},\frac{1}{3}\right\rangle\right\}.$$

10. We have $\mathbf{u}_1 = \langle -3,1,1\rangle, \mathbf{u}_2 = \langle 1,1,0\rangle$, and $\mathbf{u}_3 = \langle -1,4,1\rangle$. Taking $\mathbf{v}_1 = \mathbf{u}_1 = \langle -3,1,1\rangle$ and using $\mathbf{u}_2 \cdot \mathbf{v}_1 = -2$ and $\mathbf{v}_1 \cdot \mathbf{v}_1 = 11$ we obtain

$$\mathbf{v}_2 = \mathbf{u}_2 - \frac{\mathbf{u}_2 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1}\mathbf{v}_1 = \langle 1,1,0\rangle - \frac{-2}{11}\langle -3,1,1\rangle = \left\langle \frac{5}{11},\frac{13}{11},\frac{2}{11}\right\rangle.$$

Next, using $\mathbf{u}_3 \cdot \mathbf{v}_1 = 8, \mathbf{u}_3 \cdot \mathbf{v}_2 = \frac{49}{11}$, and $\mathbf{v}_2 \cdot \mathbf{v}_2 = \frac{18}{11}$, we obtain

$$\mathbf{v}_3 = \mathbf{u}_3 - \frac{\mathbf{u}_3 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1}\mathbf{v}_1 - \frac{\mathbf{u}_3 \cdot \mathbf{v}_2}{\mathbf{v}_2 \cdot \mathbf{v}_2}\mathbf{v}_2 = \langle -1,4,1\rangle - \frac{8}{11}\langle -3,1,1\rangle - \frac{49/11}{18/11}\left\langle \frac{5}{11},\frac{13}{11},\frac{2}{11}\right\rangle = \left\langle -\frac{1}{18},\frac{1}{18},-\frac{2}{9}\right\rangle.$$

Thus, an orthogonal basis is

$$B' = \left\{\langle -3,1,1\rangle, \left\langle \frac{5}{11},\frac{13}{11},\frac{2}{11}\right\rangle, \left\langle -\frac{1}{18},\frac{1}{18},-\frac{2}{9}\right\rangle\right\},$$

and an orthonormal basis is

$$B'' = \left\{\left\langle -\frac{3}{\sqrt{11}},\frac{1}{\sqrt{11}},\frac{1}{\sqrt{11}}\right\rangle, \left\langle \frac{5}{3\sqrt{22}},\frac{13}{3\sqrt{22}},\frac{2}{3\sqrt{22}}\right\rangle, \left\langle -\frac{1}{3\sqrt{2}},\frac{1}{3\sqrt{2}},\frac{4}{3\sqrt{2}}\right\rangle\right\}.$$

11. We have $\mathbf{u}_1 = \langle \frac{1}{2},\frac{1}{2},1\rangle, \mathbf{u}_2 = \langle -1,1,-\frac{1}{2}\rangle$, and $\mathbf{u}_3 = \langle -1,\frac{1}{2},1\rangle$. Taking $\mathbf{v}_1 = \mathbf{u}_1 = \langle \frac{1}{2},\frac{1}{2},1\rangle$ and using $\mathbf{u}_2 \cdot \mathbf{v}_1 = -\frac{1}{2}$ and $\mathbf{v}_1 \cdot \mathbf{v}_1 = \frac{3}{2}$ we obtain

$$\mathbf{v}_2 = \mathbf{u}_2 - \frac{\mathbf{u}_2 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1}\mathbf{v}_1 = \left\langle -1,1,-\frac{1}{2}\right\rangle - \frac{-1/2}{3/2}\left\langle \frac{1}{2},\frac{1}{2},1\right\rangle = \left\langle -\frac{5}{6},\frac{7}{6},-\frac{1}{6}\right\rangle.$$

Next, using $\mathbf{u}_3 \cdot \mathbf{v}_1 = \frac{3}{4}, \mathbf{u}_3 \cdot \mathbf{v}_2 = \frac{5}{4}$, and $\mathbf{v}_2 \cdot \mathbf{v}_2 = \frac{25}{12}$, we obtain

$$\mathbf{v}_3 = \mathbf{u}_3 - \frac{\mathbf{u}_3 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1}\mathbf{v}_1 - \frac{\mathbf{u}_3 \cdot \mathbf{v}_2}{\mathbf{v}_2 \cdot \mathbf{v}_2}\mathbf{v}_2 = \left\langle -1,\frac{1}{2},1\right\rangle - \frac{3/4}{3/2}\left\langle \frac{1}{2},\frac{1}{2},1\right\rangle - \frac{5/4}{25/12}\left\langle -\frac{5}{6},\frac{7}{6},-\frac{1}{6}\right\rangle = \left\langle -\frac{3}{4},-\frac{9}{20},\frac{3}{5}\right\rangle.$$

Thus, an orthogonal basis is

$$B' = \left\{\left\langle \frac{1}{2},\frac{1}{2},1\right\rangle, \left\langle -\frac{5}{6},\frac{7}{6},-\frac{1}{6}\right\rangle, \left\langle -\frac{3}{4},-\frac{9}{20},\frac{3}{5}\right\rangle\right\},$$

and an orthonormal basis is

$$B'' = \left\{\left\langle \frac{1}{\sqrt{6}},\frac{1}{\sqrt{6}},\frac{2}{\sqrt{6}}\right\rangle, \left\langle -\frac{1}{\sqrt{3}},\frac{7}{5\sqrt{3}},-\frac{1}{5\sqrt{3}}\right\rangle, \left\langle -\frac{1}{\sqrt{2}},-\frac{3}{5\sqrt{2}},\frac{4}{5\sqrt{2}}\right\rangle\right\}.$$

12. We have $\mathbf{u}_1 = \langle 1,1,1\rangle, \mathbf{u}_2 = \langle 9,-1,1\rangle$, and $\mathbf{u}_3 = \langle -1,4,-2\rangle$. Taking $\mathbf{v}_1 = \mathbf{u}_1 = \langle 1,1,1\rangle$ and using $\mathbf{u}_2 \cdot \mathbf{v}_1 = 9$ and $\mathbf{v}_1 \cdot \mathbf{v}_1 = 3$ we obtain

$$\mathbf{v}_2 = \mathbf{u}_2 - \frac{\mathbf{u}_2 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1}\mathbf{v}_1 = \langle 9,-1,1\rangle - \frac{9}{3}\langle 1,1,1\rangle = \langle 6,-4,-2\rangle.$$

Next, using $\mathbf{u}_3 \cdot \mathbf{v}_1 = 1, \mathbf{u}_3 \cdot \mathbf{v}_2 = -18$, and $\mathbf{v}_2 \cdot \mathbf{v}_2 = 56$, we obtain

$$\mathbf{v}_3 = \mathbf{u}_3 - \frac{\mathbf{u}_3 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1}\mathbf{v}_1 - \frac{\mathbf{u}_3 \cdot \mathbf{v}_2}{\mathbf{v}_2 \cdot \mathbf{v}_2}\mathbf{v}_2 = \langle -1, 4, -2 \rangle - \frac{1}{3}\langle 1,1,1 \rangle - \frac{-18}{56}\langle 6,-4,-2 \rangle = \left\langle \frac{25}{42}, \frac{50}{21}, -\frac{125}{42} \right\rangle.$$

Thus, an orthogonal basis is

$$B' = \left\{ \langle 1,1,1 \rangle, \langle 6,-4,-2 \rangle, \left\langle \frac{25}{42}, \frac{50}{21}, -\frac{125}{42} \right\rangle \right\},$$

and an orthonormal basis is

$$B'' = \left\{ \left\langle \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right\rangle, \left\langle \frac{3}{\sqrt{14}}, -\frac{2}{\sqrt{14}}, -\frac{1}{\sqrt{14}} \right\rangle, \left\langle \frac{1}{\sqrt{42}}, \frac{4}{\sqrt{42}}, -\frac{5}{\sqrt{42}} \right\rangle \right\}.$$

13. We have $\mathbf{u}_1 = \langle 1,5,2 \rangle$, and $\mathbf{u}_2 = \langle -2,1,1 \rangle$. Taking $\mathbf{v}_1 = \mathbf{u}_1 = \langle 1,5,2 \rangle$ and using $\mathbf{u}_2 \cdot \mathbf{v}_1 = 5$ and $\mathbf{v}_1 \cdot \mathbf{v}_1 = 30$ we obtain

$$\mathbf{v}_2 = \mathbf{u}_2 - \frac{\mathbf{u}_2 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1}\mathbf{v}_1 = \langle -2,1,1 \rangle - \frac{5}{30}\langle 1,5,2 \rangle = \left\langle -\frac{13}{6}, \frac{1}{6}, \frac{2}{3} \right\rangle.$$

Thus, an orthogonal basis is $B' = \left\{ \langle 1,5,2 \rangle, \left\langle -\frac{13}{6}, \frac{1}{6}, \frac{2}{3} \right\rangle \right\}$, and an orthonormal basis is

$$B'' = \left\{ \left\langle \frac{1}{\sqrt{30}}, \frac{5}{\sqrt{30}}, \frac{2}{\sqrt{30}} \right\rangle, \left\langle -\frac{13}{\sqrt{186}}, \frac{1}{\sqrt{186}}, \frac{4}{\sqrt{186}} \right\rangle \right\}.$$

14. We have $\mathbf{u}_1 = \langle 1,2,3 \rangle$, and $\mathbf{u}_2 = \langle 3,4,1 \rangle$. Taking $\mathbf{v}_1 = \mathbf{u}_1 = \langle 1,2,3 \rangle$ and using $\mathbf{u}_2 \cdot \mathbf{v}_1 = 14$ and $\mathbf{v}_1 \cdot \mathbf{v}_1 = 14$ we obtain

$$\mathbf{v}_2 = \mathbf{u}_2 - \frac{\mathbf{u}_2 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1}\mathbf{v}_1 = \langle 3,4,1 \rangle - \frac{14}{14}\langle 1,2,3 \rangle = \langle 2,2,-2 \rangle.$$

Thus, an orthogonal basis is $B' = \{ \langle 1,2,3 \rangle, \langle 2,2,-2 \rangle \}$, and an orthonormal basis is

$$B'' = \left\{ \left\langle \frac{1}{\sqrt{14}}, \frac{2}{\sqrt{14}}, \frac{3}{\sqrt{14}} \right\rangle, \left\langle \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}} \right\rangle \right\}.$$

15. We have $\mathbf{u}_1 = \langle 1,-1,1,-1 \rangle$, and $\mathbf{u}_2 = \langle 1,3,0,1 \rangle$. Taking $\mathbf{v}_1 = \mathbf{u}_1 = \langle 1,-1,1,-1 \rangle$ and using $\mathbf{u}_2 \cdot \mathbf{v}_1 = -3$ and $\mathbf{v}_1 \cdot \mathbf{v}_1 = 4$ we obtain

$$\mathbf{v}_2 = \mathbf{u}_2 - \frac{\mathbf{u}_2 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1}\mathbf{v}_1 = \langle 1,3,0,1 \rangle - \frac{-3}{4}\langle 1,-1,1,-1 \rangle = \left\langle \frac{7}{4}, \frac{9}{4}, \frac{3}{4}, \frac{1}{4} \right\rangle.$$

Thus, an orthogonal basis is $B' = \left\{ \langle 1,-1,1,-1 \rangle, \left\langle \frac{7}{4}, \frac{9}{4}, \frac{3}{4}, \frac{1}{4} \right\rangle \right\}$, and an orthonormal basis is

$$B'' = \left\{ \left\langle \frac{1}{2}, -\frac{1}{2}, \frac{1}{2}, -\frac{1}{2} \right\rangle, \left\langle \frac{7}{2\sqrt{35}}, \frac{9}{2\sqrt{35}}, \frac{3}{2\sqrt{35}}, \frac{1}{2\sqrt{35}} \right\rangle \right\}.$$

16. We have $\mathbf{u}_1 = \langle 4,0,2,-1 \rangle, \mathbf{u}_2 = \langle 2,1,-1,1 \rangle$, and $\mathbf{u}_3 = \langle 1,1,-1,0 \rangle$. Taking $\mathbf{v}_1 = \mathbf{u}_1 = \langle 4,0,2,-1 \rangle$ and using $\mathbf{u}_2 \cdot \mathbf{v}_1 = 5$ and $\mathbf{v}_1 \cdot \mathbf{v}_1 = 21$ we obtain

$$\mathbf{v}_2 = \mathbf{u}_2 - \frac{\mathbf{u}_2 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1}\mathbf{v}_1 = \langle 2,1,-1,1 \rangle - \frac{5}{21}\langle 4,0,2,-1 \rangle = \left\langle \frac{22}{21}, 1, -\frac{31}{21}, \frac{26}{21} \right\rangle.$$

Next, using $\mathbf{u}_3 \cdot \mathbf{v}_1 = 2, \mathbf{u}_3 \cdot \mathbf{v}_2 = \frac{74}{21}$, and $\mathbf{v}_2 \cdot \mathbf{v}_2 = \frac{122}{21}$, we obtain

$$\begin{aligned}\mathbf{v}_3 &= \mathbf{u}_3 - \frac{\mathbf{u}_3 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1}\mathbf{v}_1 - \frac{\mathbf{u}_3 \cdot \mathbf{v}_2}{\mathbf{v}_2 \cdot \mathbf{v}_2}\mathbf{v}_2 \\ &= \langle 1,1,-1,0 \rangle - \frac{2}{21}\langle 4,0,2,-1 \rangle - \frac{74/21}{122/21}\left\langle \frac{22}{21}, 1, -\frac{31}{21}, \frac{26}{21} \right\rangle = \left\langle -\frac{1}{61}, \frac{24}{61}, -\frac{18}{61}, -\frac{40}{61} \right\rangle.\end{aligned}$$

Thus, an orthogonal basis is

$$B' = \left\{ \langle 4,0,2,-1 \rangle, \left\langle \frac{22}{21}, 1, -\frac{31}{21}, \frac{26}{21} \right\rangle, \left\langle -\frac{1}{61}, \frac{24}{61}, -\frac{18}{61}, -\frac{40}{61} \right\rangle \right\},$$

and an orthonormal basis is

$$B'' = \left\{ \left\langle \frac{4}{\sqrt{21}}, 0, \frac{2}{\sqrt{21}}, -\frac{1}{\sqrt{21}} \right\rangle, \left\langle \frac{22}{\sqrt{2562}}, \frac{21}{\sqrt{2562}}, -\frac{31}{\sqrt{2562}}, \frac{26}{\sqrt{2562}} \right\rangle, \right.$$

$$\left. \left\langle -\frac{1}{\sqrt{2501}}, \frac{24}{\sqrt{2501}}, -\frac{18}{\sqrt{2501}}, -\frac{40}{\sqrt{2501}} \right\rangle \right\}.$$

17. We have $u_1 = 1, u_2 = x$, and $u_3 = x^2$. Taking $v_1 = u_1 = 1$ and using

$$(u_2, v_1) = \int_{-1}^{1} 1 \cdot x^2 \, dx = 0 \quad \text{and} \quad (v_1, v_1) = \int_{-1}^{1} x \cdot x \, dx = 2$$

we obtain

$$v_2 = u_2 - \frac{(u_2, v_1)}{(v_1, v_1)} v_1 = x - \frac{0}{2} x = x.$$

Next, using

$$(u_3, v_1) = \int_{-1}^{1} x^2 \cdot 1 \, dx = \frac{2}{3}, \quad (u_3, v_2) = \int_{-1}^{1} x^2 \cdot x \, dx = 0, \quad \text{and} \quad (v_2, v_2) = \int_{-1}^{1} x \cdot x \, dx = \frac{2}{3},$$

we obtain

$$v_3 = u_3 - \frac{(u_3, v_1)}{(v_1, v_1)} v_1 - \frac{(u_3, v_2)}{(v_2, v_2)} v_2 = x^2 - \frac{2/3}{2} 1 - \frac{0}{2/3} x = x^2 - \frac{1}{3}.$$

Thus, an orthogonal basis is $B' = \left\{1, x, x^2 - \frac{1}{3}\right\}$.

18. We have $u_1 = x^2 - x, u_2 = x^2 + 1$, and $u_3 = 1 - x^2$. Taking $v_1 = u_1 = x^2 - x$ and using

$$(u_2, v_1) = \int_{-1}^{1} (x^2 + 1)(x^2 - x) dx = \frac{16}{15} \quad \text{and} \quad (v_1, v_1) = \int_{-1}^{1} (x^2 - x)(x^2 - x) dx = \frac{16}{15}$$

we obtain

$$v_2 = u_2 - \frac{(u_2, v_1)}{(v_1, v_1)} v_1 = x^2 + 1 - \frac{16/15}{16/15} (x^2 - x) = x + 1.$$

Next, using

$$(u_3, v_1) = \int_{-1}^{1} (1 - x^2)(x^2 - x) dx = \frac{4}{15}, \quad (u_3, v_2) = \int_{-1}^{1} (1 - x^2)(x + 1) dx = \frac{4}{3},$$

and

$$(v_2, v_2) = \int_{-1}^{1} (x + 1)(x + 1) dx = \frac{8}{3},$$

we obtain

$$v_3 = u_3 - \frac{(u_3, v_1)}{(v_1, v_1)} v_1 - \frac{(u_3, v_2)}{(v_2, v_2)} v_2 = 1 - x^2 - \frac{4/15}{16/15}(x^2 - x) - \frac{4/3}{8/3}(x + 1) = -\frac{5}{4} x^3 - \frac{1}{4} x + \frac{1}{2}.$$

Thus, an orthogonal basis is $B' = \left\{x^2 - x, x + 1, -\frac{5}{4} x^3 - \frac{1}{4} x + \frac{1}{2}\right\}$.

19. Using the solution of Problem 17 and computing

$$||v_1||^2 = (v_1, v_1) = \int_{-1}^{1} 1 \cdot 1 \, dx = 2, \quad ||v_2||^2 = (v_2, v_2) = \int_{-1}^{1} x \cdot x \, dx = \frac{2}{3},$$

and

$$||v_3||^2 = (v_3, v_3) = \int_{-1}^{1} \left(x^2 - \frac{1}{3}\right) \left(x^2 - \frac{1}{3}\right) dx = \frac{8}{45},$$

we see that an orthonormal basis is

$$B'' = \left\{ \frac{1}{\sqrt{2}}, \frac{x}{\sqrt{2/3}}, \frac{x^2 - 1/3}{\sqrt{8/45}} \right\} = \left\{ \frac{1}{\sqrt{2}}, \frac{3}{\sqrt{6}}\,x, \frac{15}{2\sqrt{10}}\left(x^2 - \frac{1}{3}\right) \right\}.$$

20. Using the solution of Problem 18 and computing

$$||v_1||^2 = (v_1, v_1) = \int_{-1}^{1} (x^2 - x)(x^2 - x)dx = \frac{16}{15}, \quad ||v_2||^2 = (v_2, v_2) = \int_{-1}^{1} (x+1)(x+1)dx = \frac{8}{3},$$

and

$$||v_3||^2 = (v_3, v_3) = \int_{-1}^{1} \left(-\frac{5}{4}\,x^3 - \frac{1}{4}\,x + \frac{1}{2} \right) \left(-\frac{5}{4}\,x^3 - \frac{1}{4}\,x + \frac{1}{2} \right) dx = \frac{1}{3},$$

we see that an orthonormal basis is

$$B'' = \left\{ \frac{\sqrt{15}}{4}\,(x^2 - x), \frac{3}{2\sqrt{6}}\,(x+1), \frac{\sqrt{3}}{4}\,(-5x^2 - x + 2) \right\}.$$

21. Using $w_1 = 1/\sqrt{2}, w_2 = 3x/\sqrt{6}$, and $w_3 = (15/2\sqrt{10})(x^2 - 1/3)$, and computing

$$(p, w_1) = \int_{-1}^{1} (9x^2 - 6x + 5)\,\frac{1}{\sqrt{2}}\,dx = 8\sqrt{2},$$

$$(p, w_2) = \int_{-1}^{1} (9x^2 - 6x + 5)\,\frac{3}{\sqrt{6}}\,x\,dx = -2\sqrt{6}$$

$$(p, w_3) = \int_{-1}^{1} (9x^2 - 6x + 5)\left[\frac{15}{2\sqrt{10}}\left(x^2 - \frac{1}{3}\right)\right] dx = \frac{12}{\sqrt{10}},$$

we find from Theorem 7.5

$$p(x) = 9x^2 - 6x + 5 = (p, w_1)w_1 + (p, w_2)w_2 + (p, w_3)w_3 = 8\sqrt{2}\,w_1 - 2\sqrt{6}\,w_2 + \frac{12}{\sqrt{10}}\,w_3.$$

22. Using $w_1 = (\sqrt{15}/4)(x^2 - x), w_2 = (3/2\sqrt{6})(x+1)$, and $w_3 = -(\sqrt{3}/4)(5x^2 + x - 2)$, and computing

$$(p, w_1) = \int_{-1}^{1} (9x^2 - 6x + 5)\left[\frac{\sqrt{15}}{4}\,(x^2 - x)\right] dx = \frac{41}{\sqrt{15}},$$

$$(p, w_2) = \int_{-1}^{1} (9x^2 - 6x + 5)\left[\frac{3}{2\sqrt{6}}\,(x+1)\right] dx = 3\sqrt{6}$$

$$(p, w_3) = \int_{-1}^{1} (9x^2 - 6x + 5)\left[-\frac{\sqrt{3}}{4}\,(5x^2 + x - 2)\right] dx = \frac{1}{\sqrt{3}},$$

we find from Theorem 7.5

$$p(x) = 9x^2 - 6x + 5 = (p, w_1)w_1 + (p, w_2)w_2 + (p, w_3)w_3 = \frac{41}{\sqrt{15}}\,w_1 + 3\sqrt{6}\,w_2 + \frac{1}{\sqrt{3}}\,w_3.$$

23. Since $\mathbf{u}_3$ depends on $\mathbf{u}_1$ and $\mathbf{u}_2$ we would expect the Gram-Schmidt process to yield a pair of orthogonal vectors $\mathbf{v}_1$ and $\mathbf{v}_2$, with a third vector $\mathbf{v}_3$ that is $\mathbf{0}$. This is because $\mathbf{u}_3$ lies in the subspace W_2 of R^3 spanned by $\mathbf{u}_1$ and $\mathbf{u}_2$, and hence the projection of $\mathbf{u}_3$ onto W_2 is $\mathbf{u}_3$ itself. In other words,

$$\mathbf{u}_3 = \mathrm{proj}_{W_3}\mathbf{u}_3 = \frac{\mathbf{u}_3 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1}\,\mathbf{v}_1 + \frac{\mathbf{u}_3 \cdot \mathbf{v}_2}{\mathbf{v}_2 \cdot \mathbf{v}_2}\,\mathbf{v}_2 \quad \text{so} \quad \mathbf{v}_3 = \mathbf{u}_3 - \frac{\mathbf{u}_3 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1}\,\mathbf{v}_1 + \frac{\mathbf{u}_3 \cdot \mathbf{v}_2}{\mathbf{v}_2 \cdot \mathbf{v}_2}\,\mathbf{v}_2 = \mathbf{0}.$$

To carry out the orthogonalization process we take $\mathbf{v}_1 = \mathbf{u}_1 = \langle 1, 1, 3 \rangle$. Then, using $\mathbf{u}_2 \cdot \mathbf{v}_1 = 8$ and $\mathbf{v}_1 \cdot \mathbf{v}_1 = 11$ we obtain

$$\mathbf{v}_2 = \mathbf{u}_2 - \frac{\mathbf{u}_2 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1}\,\mathbf{v}_1 = \langle 1, 4, 1 \rangle - \frac{8}{11}\langle 1, 1, 3 \rangle = \left\langle \frac{3}{11}, \frac{36}{11}, -\frac{13}{11} \right\rangle.$$

Next, using $\mathbf{u}_3 \cdot \mathbf{v}_1 = 2, \mathbf{u}_3 \cdot \mathbf{v}_2 = \frac{402}{11}$, and $\mathbf{v}_2 \cdot \mathbf{v}_2 = \frac{134}{11}$, we obtain

$$\mathbf{v}_3 = \mathbf{u}_3 - \frac{\mathbf{u}_3 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1}\mathbf{v}_1 - \frac{\mathbf{u}_3 \cdot \mathbf{v}_2}{\mathbf{v}_2 \cdot \mathbf{v}_2}\mathbf{v}_2 = \langle 1, 10, -3\rangle - \frac{2}{11}\langle 1, 1, 3\rangle - \frac{402/11}{134/11}\left\langle \frac{3}{11}, \frac{36}{11}, -\frac{13}{11}\right\rangle = \langle 0, 0, 0\rangle .$$

In this case $\{\mathbf{v}_1, \mathbf{v}_2\} = \{\langle 1, 1, 3\rangle, \langle \frac{3}{11}, \frac{36}{11}, -\frac{13}{11}\rangle\}$ is an orthogonal subset of R^3 containing the third vector $\mathbf{u}_3 = \langle 1, 10, -3\rangle$.

CHAPTER 7 REVIEW EXERCISES

1. True

2. False; the points must be non-collinear.

3. False; since a normal to the plane is $\langle 2, 3, -4\rangle$ which is not a multiple of the direction vector $\langle 5, -2, 1\rangle$ of the line.

4. True **5.** True **6.** True **7.** True **8.** True **9.** True

10. True; since $\mathbf{a} \times \mathbf{b}$ and $\mathbf{c} \times \mathbf{d}$ are both normal to the plane and hence parallel (unless $\mathbf{a} \times \mathbf{b} = \mathbf{0}$ or $\mathbf{c} \times \mathbf{d} = \mathbf{0}$.)

11. $9\mathbf{i} + 2\mathbf{j} + 2\mathbf{k}$

12. orthogonal

13. $-5(\mathbf{k} \times \mathbf{j}) = -5(-\mathbf{i}) = 5\mathbf{i}$

14. $\mathbf{i} \cdot (\mathbf{i} \times \mathbf{j}) = \mathbf{i} \times \mathbf{k} = \mathbf{0}$

15. $\sqrt{(-12)^2 + 4^2 + 6^2} = 14$

16. $(-1 - 20)\mathbf{i} - (-2 - 0)\mathbf{j} + (8 - 0)\mathbf{k} = -21\mathbf{i} + 2\mathbf{j} + 8\mathbf{k}$

17. $-6\mathbf{i} + \mathbf{j} - 7\mathbf{k}$

18. The coordinates of $(1, -2, -10)$ satisfy the given equation.

19. Writing the line in parametric form, we have $x = 1 + t$, $y = -2 + 3t$, $z = -1 + 2t$. Substituting into the equation of the plane yields $(1 + t) + 2(-2 + 3t) - (-1 + 2t) = 13$ or $t = 3$. Thus, the point of intersection is $x = 1 + 3 = 4$, $y = -2 + 3(3) = 7$, $z = -1 + 2(3) = 5$, or $(4, 7, 5)$.

20. $|\mathbf{a}| = \sqrt{4^2 + 3^2 + (-5)^2} = 5\sqrt{2}\,;\;\; \mathbf{u} = -\dfrac{1}{5\sqrt{2}}(4\mathbf{i} + 3\mathbf{j} - 5\mathbf{k}) = -\dfrac{4}{5\sqrt{2}}\mathbf{i} - \dfrac{3}{5\sqrt{2}}\mathbf{j} + \dfrac{1}{\sqrt{2}}\mathbf{k}$

21. $x_2 - 2 = 3,\;\; x_2 = 5;\;\; y_2 - 1 = 5,\;\; y_2 = 6;\;\; z_2 - 7 = -4,\;\; z_2 = 3;\;\; P_2 = (5, 6, 3)$

22. $(5, 1/2, 5/2)$

23. $(7.2)(10)\cos 135° = -36\sqrt{2}$

24. $2\mathbf{b} = \langle -2, 4, 2\rangle;\;\; 4\mathbf{c} = \langle 0, -8, 8\rangle;\;\; \mathbf{a} \cdot (2\mathbf{b} + 4\mathbf{c}) = \langle 3, 1, 0\rangle \cdot \langle -2, -4, 10\rangle = -10$

25. $12,\ -8,\ 6$

26. $\cos\theta = \dfrac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}||\mathbf{b}|} = \dfrac{1}{\sqrt{2}\,\sqrt{2}} = \dfrac{1}{2};\ \theta = 60°$

27. $A = \dfrac{1}{2}|5\mathbf{i} - 4\mathbf{j} - 7\mathbf{k}| = \dfrac{3\sqrt{10}}{2}$

28. From $3(x-3)+0(y-6)+(1)(z-(-2))=0$ we obtain $3x+z=7$.

29. $|-5-(-3)|=2$

30. parallel: $-2c=5,\ \ c=-5/2$; orthogonal: $1(-2)+3(-6)+c(5)=0,\ \ c=4$

31. $\mathbf{a}\times\mathbf{b}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\1&1&0\\1&-2&1\end{vmatrix}=\begin{vmatrix}1&0\\-2&1\end{vmatrix}\mathbf{i}-\begin{vmatrix}1&0\\1&1\end{vmatrix}\mathbf{j}+\begin{vmatrix}1&1\\1&-2\end{vmatrix}\mathbf{k}=\mathbf{i}-\mathbf{j}-3\mathbf{k}$ A unit vector perpendicular to both $\mathbf{a}$ and $\mathbf{b}$ is

$$\frac{\mathbf{a}\times\mathbf{b}}{\|\mathbf{a}\times\mathbf{b}\|}=\frac{1}{\sqrt{1+1+9}}(\mathbf{i}-\mathbf{j}-3\mathbf{k})=\frac{1}{\sqrt{11}}\mathbf{i}-\frac{1}{\sqrt{11}}\mathbf{j}-\frac{3}{\sqrt{11}}\mathbf{k}.$$

32. $\|\mathbf{a}\|=\sqrt{1/4+1/4+1/6}=\dfrac{3}{4}$; $\cos\alpha=\dfrac{1/2}{3/4}=\dfrac{2}{3}$, $\alpha\approx 48.19°$; $\cos\beta=\dfrac{1/2}{3/4}=\dfrac{2}{3}$, $\beta\approx 48.19°$; $\cos\gamma=\dfrac{-1/4}{3/4}=-\dfrac{1}{3}$, $\gamma\approx 109.47°$

33. $\text{comp}_{\mathbf{b}}\mathbf{a}=\mathbf{a}\cdot\mathbf{b}/\|\mathbf{b}\|=\langle 1,2,-2\rangle\cdot\langle 4,3,0\rangle/5=2$

34. $\text{comp}_{\mathbf{a}}\mathbf{b}=\mathbf{b}\cdot\mathbf{a}/\|\mathbf{a}\|=\langle 4,3,0\rangle\cdot\langle 1,2,-2\rangle/3=10/3$
$\text{proj}_{\mathbf{a}}\mathbf{b}=(\text{comp}_{\mathbf{a}}\mathbf{b})\mathbf{a}/\|\mathbf{a}\|=(10/3)\langle 1,2,-2\rangle/3=\langle 10/9,20/9,-20/9\rangle$

35. $\mathbf{a}+\mathbf{b}=\langle 1,2,-2\rangle+\langle 4,3,0\rangle=\langle 5,5,-2\rangle$
$\text{comp}_{\mathbf{a}}(\mathbf{a}+\mathbf{b})=(\mathbf{a}+\mathbf{b})\cdot\mathbf{a}/\sqrt{1+4+4}=\frac{1}{3}(\mathbf{a}\cdot\mathbf{a}+\mathbf{b}\cdot\mathbf{a})=\frac{1}{3}[(1+4+4)+(4+6+0)]=\frac{19}{3}$
$\text{proj}_{\mathbf{a}}(\mathbf{a}+\mathbf{b})=[\text{comp}_{\mathbf{a}}(\mathbf{a}+\mathbf{b})](\mathbf{a}/\|\mathbf{a}\|)=\frac{19}{3}\langle\frac{1}{3},\frac{2}{3},-\frac{2}{3}\rangle=\langle\frac{19}{9},\frac{38}{9},-\frac{38}{9}\rangle$

36. $\mathbf{a}-\mathbf{b}=\langle 1,2,-2\rangle-\langle 4,3,0\rangle=\langle -3,-1,-2\rangle$
$\text{comp}_{\mathbf{b}}(\mathbf{a}-\mathbf{b})=(\mathbf{a}-\mathbf{b})\cdot\mathbf{b}/\sqrt{16+9}=\frac{1}{5}(\mathbf{a}\cdot\mathbf{b}-\mathbf{b}\cdot\mathbf{b})=\frac{1}{5}[(4+6+0)-(16+9)]=-3$
$\text{proj}_{\mathbf{b}}(\mathbf{a}-\mathbf{b})=[\text{comp}_{\mathbf{b}}(\mathbf{a}-\mathbf{b})](\mathbf{b}/\|\mathbf{b}\|)=-3\langle\frac{4}{5},\frac{3}{5},0\rangle=\langle -\frac{12}{5},-\frac{9}{5},0\rangle$

37. Let $\mathbf{a}=\langle a,b,c\rangle$ and $\mathbf{r}=\langle x,y,z\rangle$. Then

(a) $(\mathbf{r}-\mathbf{a})\cdot\mathbf{r}=\langle x-a,y-b,z-c\rangle\cdot\langle x,y,z\rangle=x^2-ax+y^2-by+z^2-zc=0$ implies

$$(x-\frac{a}{2})^2+(y-\frac{b}{2})^2+(z-\frac{c}{2})^2=\frac{a^2+b^2+c^2}{4}.$$ The surface is a sphere.

(b) $(\mathbf{r}-\mathbf{a})\cdot\mathbf{a}=\langle x-a,y-b,z-c\rangle\cdot\langle a,b,c\rangle=a(x-a)+b(y-b)+c(z-c)=0$
The surface is a plane.

38. $\langle 4,2,-2\rangle-\langle 2,4,-3\rangle=\langle 2,-2,1\rangle$; $\langle 2,4,-3\rangle-\langle 6,7,-5\rangle=\langle -4,-3,2\rangle$; $\langle 2,-2,1\rangle\cdot\langle -4,-3,2\rangle=0$
The points are the vertices of a right triangle.

39. A direction vector of the given line is $\langle 4,-2,6\rangle$. A parallel line containing $(7,3,-5)$ is $(x-7)/4=(y-3)/(-2)=(z+5)/6$.

40. A normal to the plane is $\langle 8,3,-4\rangle$. The line with this direction vector and through $(5,-9,3)$ is $x=5+8t$, $y=-9+3t$, $z=3-4t$.

41. The direction vectors are $\langle -2,3,1\rangle$ and $\langle 2,1,1\rangle$. Since $\langle -2,3,1\rangle\cdot\langle 2,1,1\rangle=0$, the lines are orthogonal. Solving $1-2t=x=1+2s$, $3t=y=-4+s$, we obtain $t=-1$ and $s=1$. The point $(3,-3,0)$ obtained by letting $t=-1$ and $s=1$ is common to the two lines, so they do intersect.

42. Vectors in the plane are $\langle 2,3,1\rangle$ and $\langle 1,0,2\rangle$. A normal vector is $\langle 2,3,1\rangle\times\langle 1,0,2\rangle=\langle 6,-3,-3\rangle=3\langle 2,-1,-1\rangle$. An equation of the plane is $2x-y-z=0$

43. The lines are parallel with direction vector $\langle 1,4,-2\rangle$. Since $(0,0,0)$ is on the first line and $(1,1,3)$ is on the second line, the vector $\langle 1,1,3\rangle$ is in the plane. A normal vector to the plane is thus $\langle 1,4,-2\rangle\times\langle 1,1,3\rangle=\langle 14,-5,-3\rangle$. An equation of the plane is $14x-5y-3z=0$.

44. Letting $z = t$ in the equations of the plane and solving $-x + y = 4 + 8t$, $3x - y = -2t$, we obtain $x = 2 + 3t$, $y = 6 + 11t$, $z = t$. Thus, a normal to the plane is $\langle 3, 11, 1\rangle$ and an equation of the plane is

$$3(x-1) + 11(y-7) + (z+1) = 0 \quad \text{or} \quad 3x + 11y + z = 79.$$

45. $\mathbf{F} = 10\dfrac{\mathbf{a}}{\|\mathbf{a}\|} = \dfrac{10}{\sqrt{2}}(\mathbf{i}+\mathbf{j}) = 5\sqrt{2}\,\mathbf{i} + 5\sqrt{2}\,\mathbf{j}$; $\mathbf{d} = \langle 7,4,0\rangle - \langle 4,1,0\rangle = 3\mathbf{i} + 3\mathbf{j}$

$W = \mathbf{F}\cdot\mathbf{d} = 15\sqrt{2} + 15\sqrt{2} = 30\sqrt{2}$ N-m

46. $\mathbf{F} = 5\sqrt{2}\,\mathbf{i} + 5\sqrt{2}\,\mathbf{j} + 50\mathbf{i} = (5\sqrt{2}+50)\mathbf{i} + 5\sqrt{2}\,\mathbf{j}$; $\mathbf{d} = 3\mathbf{i} + 3\mathbf{j}$

$W = 15\sqrt{2} + 150 + 15\sqrt{2} = 30\sqrt{2} + 150$ N-m ≈ 192.4 N-m

47. Since $\mathbf{F}_2 = 200(\mathbf{i}+\mathbf{j})/\sqrt{2} = 100\sqrt{2}\,\mathbf{i} + 100\sqrt{2}\,\mathbf{j}$, $\mathbf{F}_3 = \mathbf{F}_2 - \mathbf{F}_1 = (100\sqrt{2} - 200)\mathbf{i} + 100\sqrt{2}\,\mathbf{j}$ and

$\|\mathbf{F}_3\| = \sqrt{(100\sqrt{2}-200)^2 + (100\sqrt{2})^2} = 200\sqrt{2-\sqrt{2}} \approx 153$ lb.

48. Let $\|\mathbf{F}_1\| = F_1$ and $\|\mathbf{F}_2\| = F_2$. Then $\mathbf{F}_1 = F_1[(\cos 45°)\mathbf{i} + (\sin 45°)\mathbf{j}]$ and $\mathbf{F}_2 = F_2[(\cos 120°)\mathbf{i} + (\sin 120°)\mathbf{j}]$, or $\mathbf{F}_1 = F_1(\frac{1}{\sqrt{2}}\mathbf{i} + \frac{1}{\sqrt{2}}\mathbf{j})$ and $\mathbf{F}_2 = F_2(-\frac{1}{2}\mathbf{i} + \frac{\sqrt{3}}{2}\mathbf{j})$. Since $\mathbf{w} + \mathbf{F}_1 + \mathbf{F}_2 = \mathbf{0}$,

$$F_1\Big(\frac{1}{\sqrt{2}}\mathbf{i} + \frac{1}{\sqrt{2}}\mathbf{j}\Big) + F_2\Big(-\frac{1}{2}\mathbf{i} + \frac{\sqrt{3}}{2}\mathbf{j}\Big) = 50\mathbf{j}, \qquad \Big(\frac{1}{\sqrt{2}}F_1 - \frac{1}{2}F_2\Big)\mathbf{i} + \Big(\frac{1}{\sqrt{2}}F_1 + \frac{\sqrt{3}}{2}F_2\Big)\mathbf{j} = 50\mathbf{j}$$

and

$$\frac{1}{\sqrt{2}}F_1 - \frac{1}{2}F_2 = 0, \qquad \frac{1}{\sqrt{2}}F_1 + \frac{\sqrt{3}}{2}F_2 = 50.$$

Solving, we obtain $F_1 = 25(\sqrt{6} - \sqrt{2}\,) \approx 25.9$ lb and $F_2 = 50(\sqrt{3} - 1) \approx 36.6$ lb.

49. Not a vector space. Axiom (**viii**) is not satisfied.

50. The vectors are linearly independent. The only solution of the system

$$c_1 = 0, \qquad c_1 + 2c_2 + c_3 = 0, \qquad 2c_1 + 3c_2 - c_3 = 0$$

is $c_1 = 0$, $c_2 = 0$, $c_3 = 0$.

51. Let p_1 and p_2 be in P_n such that $\dfrac{d^2p_1}{dx^2} = 0$ and $\dfrac{d^2p_2}{dx^2} = 0$. Since

$$0 = \frac{d^2p_1}{dx^2} + \frac{d^2p_2}{dx^2} = \frac{d^2}{dx^2}(p_1 + p_2) \quad \text{and} \quad 0 = k\,\frac{d^2p_1}{dx^2} = \frac{d^2}{dx^2}(kp_1)$$

we conclude that the set of polynomials with the given property is a subspace of P_n. A basis for the subspace is 1, x.

52. The intersection $W_1 \cap W_2$ is a subspace of V. If x and y are in $W_1 \cap W_2$ then $\mathbf{x}$ and $\mathbf{y}$ are in each subspace and so $\mathbf{x} + \mathbf{y}$ is in each subspace. That is, $\mathbf{x} + \mathbf{y}$ is in $W_1 \cap W_2$. Similarly, if $\mathbf{x}$ is in $W_1 \cap W_2$ then $\mathbf{x}$ is in each subspace and so $k\mathbf{x}$ is in each subspace. That is, $k\mathbf{x}$ is in $W_1 \cap W_2$ for any scalar k.

The union $W_1 \cup W_2$ is generally not a subspace. For example, $W_1 = \{\langle x, y\rangle \mid y = x\}$ and $W_2 = \{\langle x, y\rangle \mid y = 2x\}$ are subspaces of R^2. Now $\langle 1, 1\rangle$ is in W_1 and $\langle 1, 2\rangle$ is in W_2 but $\langle 1,1\rangle + \langle 1,2\rangle = \langle 2,3\rangle$ is not in $W_1 \cup W_2$.